

Figure 3.40 Speed-feedback filter

From the motor equations and the above set of definitions, the equations of the motor are written as

$$\dot{x}_1 = -\frac{R_a}{L_a}x_1 - \frac{M}{L_a}x_2x_3 + \frac{V_a}{L_a} \quad (3.136)$$

$$\dot{x}_2 = -\frac{R_f}{L_f}x_2 + \frac{V_f}{L_f} \quad (3.137)$$

$$\dot{x}_3 = \frac{M}{J}x_1x_2 - \frac{B}{J}x_3 - \frac{T_l}{J} \quad (3.138)$$

3.11.2 Filter in the Speed-Feedback Loop

Figure 3.40 shows the speed-feedback filter. The transfer function of the filter and tachogenerator can be represented as

$$G_\omega(s) = \frac{\omega_{mr}(s)}{\omega_m(s)} = \frac{H_\omega}{1 + sT_\omega} \quad (3.139)$$

In time domain, this can be rearranged by letting

$$x_4 = \omega_{mr} \quad (3.140)$$

resulting in

$$\dot{x}_4 = \frac{1}{T_\omega} (H_\omega x_3 - x_4) \quad (3.141)$$

The state variable x_4 forms one of the inputs to the speed error/controller block, and that is considered next.

3.11.3 Speed Controller

The speed-controller block diagram is shown in Figure 3.41. The transfer function of the speed controller considered in the present analysis is a proportional-plus-integral controller, given as

$$G_s(s) = K_{ps} + \frac{K_{is}}{s} \quad (3.142)$$

The state diagram of this block is shown in Figure 3.42. Letting

$$\dot{x}_5 = \omega_r^* - \omega_{mr} = \omega_r^* - x_4 \quad (3.143)$$

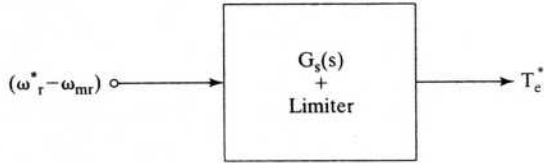


Figure 3.41 Speed-controller block diagram

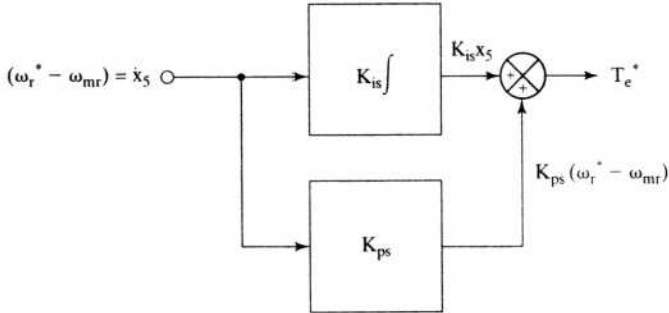


Figure 3.42 State diagram of the speed controller

the torque command signal is derived as

$$T_e^* = K_{ps}(\omega_r^* - x_4) + K_{is}x_5 \tag{3.144}$$

In order to maintain the drive system in the safe operating region, the torque reference is limited to allowable maximum limits determined by the converter and motor peak capabilities. In this case, let that be $+T_{max}$. This torque reference limit is integrated into the simulation as

$$0 \leq T_e^* \leq +T_{max} \tag{3.145}$$

3.11.4 Current-Reference Generator

The current reference is derived from the torque reference by using the relationship

$$i_a^* = \frac{T_e^*}{M i_f} = \frac{K_{ps}}{M} \cdot \frac{\omega_r^*}{x_2} - \frac{K_{ps}}{M} \cdot \frac{x_4}{x_2} + \frac{K_{is}}{M} \cdot \frac{x_5}{x_2} \tag{3.146}$$

3.11.5 Current Controller

The current controller is of a proportional-plus-integral type with limiter. The transfer function of the current controller is

$$G_i(s) = K_{pi} + \frac{K_{ii}}{s} \tag{3.147}$$

Similar to the speed-controller equations, the current-controller equations are

$$\dot{x}_6 = (i_a^* - H_c i_a) = (i_a^* - H_c x_1) \tag{3.148}$$

and

$$v_c = K_{pi}(i_a^* - H_c x_1) + K_{ii} x_6 \quad (3.149)$$

This control voltage v_c has to be limited to a value V_{cm} corresponding to I_{max} , given in the present case as

$$V_{cm} \equiv H_c I_{max} \quad (3.150)$$

where I_{max} is the maximum allowable current in the motor and converter. Hence, the limiter is prescribed as

$$0 \leq v_c \leq V_{cm} \quad (3.151)$$

The lower limit for v_c is made zero: the current in the phase-controlled rectifier cannot be reversed. In the case of a four-quadrant dc motor drive, that limit can be $-V_{cm}$.

Linearizing controller: The linearization of the output-voltage to input-control signal is accomplished by the following:

$$\alpha = \cos^{-1}(v_c/V_{cm}) \quad (3.152)$$

It could be seen that, when control-signal command goes maximum for maximum positive current error, the result is the triggering angle, α , of zero. That provides the maximum voltage to the armature of the dc motor, building up the armature current and reducing the current error.

Bridge converter: The bridge converter for continuous current can be modeled with its delay as

$$V_a = \sqrt{2}V \sin(\omega_s t + \pi/3 + \alpha) \quad (3.153)$$

Note that, for every 60 degrees, $\omega_s t$ rolls over, and α is updated at the beginning of each 60 electrical degrees. This can be incorporated into the simulation as

$$\left. \begin{array}{l} \omega_s = 2\pi f_s \\ \omega_s t > \frac{\pi}{3}, \omega_s t = 0 \\ \text{When } \omega_s t = 0, \text{ update } \alpha \\ \text{and maintain the same } \alpha \\ \text{until } \omega_s t = \pi/3 \end{array} \right\} \quad (3.154)$$

3.11.6 Flowchart for Simulation

A flowchart for the digital computer simulation of the one-quadrant dc drive is shown in Figure 3.43. The phase-controlled converter modeling for variations of α greater than 60 degrees has to be considered; accordingly, the algorithm has to be embedded in the software. For precise prediction of SCR and motor-voltage

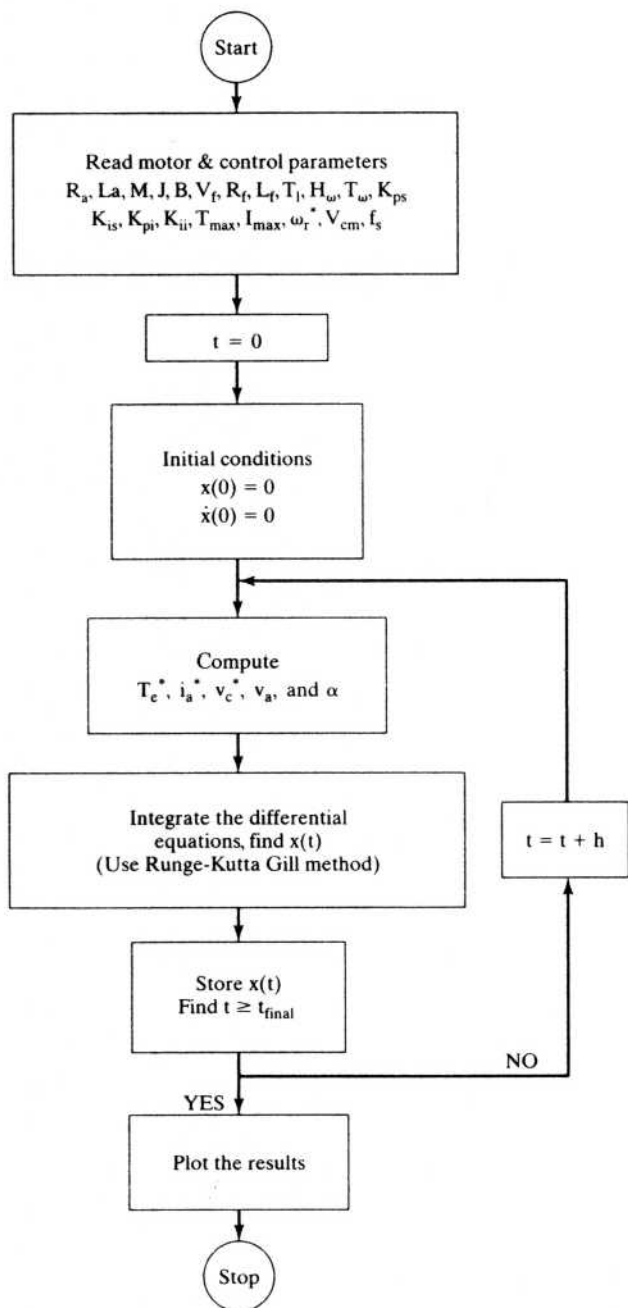


Figure 3.43 Flowchart for the simulation of a single-quadrant phase-controlled dc motor drive

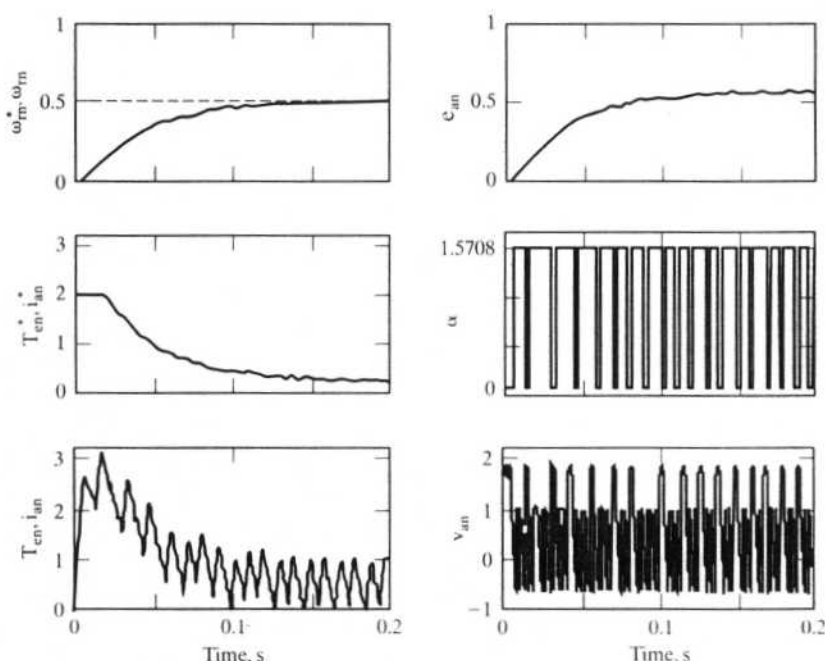


Figure 3.44 Simulation results of armature-controlled dc motor drive with 3-phase SCR converters

waveforms to validate the design of the converter, the use of the right algorithm is required.

In the case of discontinuous armature current, note that a nonzero armature current exists for the conduction duration, which will be less than 60 electrical degrees. For the remaining part of the cycle of 60 electrical degrees duration, the armature current is zero; accordingly, the algorithm has to be modified.

3.11.7 Simulation Results

A set of typical simulation results for a step change in speed reference from standstill to 0.5 p.u. with rated field current is shown in Figure 3.44. The speed error is maximum at the start, resulting in a maximum-torque reference, which is limited to 2 p.u. As the speed approaches the set value, the torque reference decreases. The average armature current settles to 0.25 p.u., to generate the electromagnetic torque to counter the load torque, which is set at 0.25 p.u. in this simulation. That happens when the rotor speed drops below the set speed by a small measure. The triggering angle delay is switched to the extremes because of the high proportional gain in the current controller. The speed oscillations are due to high proportional gain of the speed controller. The induced emf follows the rotor speed as the field flux is maintained constant in this simulation.

3.12 HARMONICS AND ASSOCIATED PROBLEMS

The ac input supplies pure sinusoidal voltages and has no harmonic contents in it. It stands to reason that there will be no harmonic currents from the source for a linear load. The SCR control in the converter causes the load to have harmonic voltages besides the dc component. The harmonic voltages produce harmonic currents, which reflect onto the ac-source side. Since the ac source does not contain harmonic sources, the harmonic currents and voltages can be attributed to the phase-controlled converter. Therefore, the phase-controlled converter can be considered a harmonic generator. These harmonics result in heating and torque pulsations in the motor and cause resonance in the power system network. The latter is treated in this section.

3.12.1 Harmonic Resonance

The interaction of the power factor-improving capacitor bank connected to the input of the ac source or the input filters with the phase-controlled dc motor drive provides an ideal setting for resonance, because the circuit is an R, L, and C network. This section contains the resonance condition for such a network and the derivation of the natural system frequency of the power system. An illustrative case study and methods to avoid resonance are outlined.

The input currents to the phase-controlled converter are assumed to be ideally rectangular blocks of alternate polarity and of duration equal to 120 electrical degrees. These waveforms, when resolved into Fourier series, have a number of harmonics besides the dominant fundamental. The fifth and seventh are 1/5th and 1/7th of the strength of the fundamental, respectively. Most of the power-system inputs are installed with large capacitor banks for power-factor correction. The capacitor bank whose reactance is X_c comes in parallel to the impedance of the network including loads $R + jX_l$, as shown in Figure 3.45. That the circuit will resonate under the condition that $X_l = X_c$ is seen from the equivalent impedance,

$$Z_{eq} = - \frac{jX_c(R + jX_l)}{R + j(X_l - X_c)} \quad (3.155)$$

When it is in resonance, the equivalent impedance is

$$Z_{eq} = - \frac{jX_c(R + jX_l)}{R} = - \left(1 + \frac{jX_l}{R} \right) jX_c \quad (3.156)$$

If $X_l/R \gg 1$, then the equivalent impedance of the network is approximated as

$$Z_{eq} = \frac{X_l}{R} X_c \quad (3.157)$$

The n^{th} harmonic equivalent impedance is then

$$Z_{eqn} = \left(\frac{nX_l}{R} \right) \left(\frac{X_c}{n} \right) = \frac{X_l}{R} X_c = Z_{eq} \quad (3.158)$$

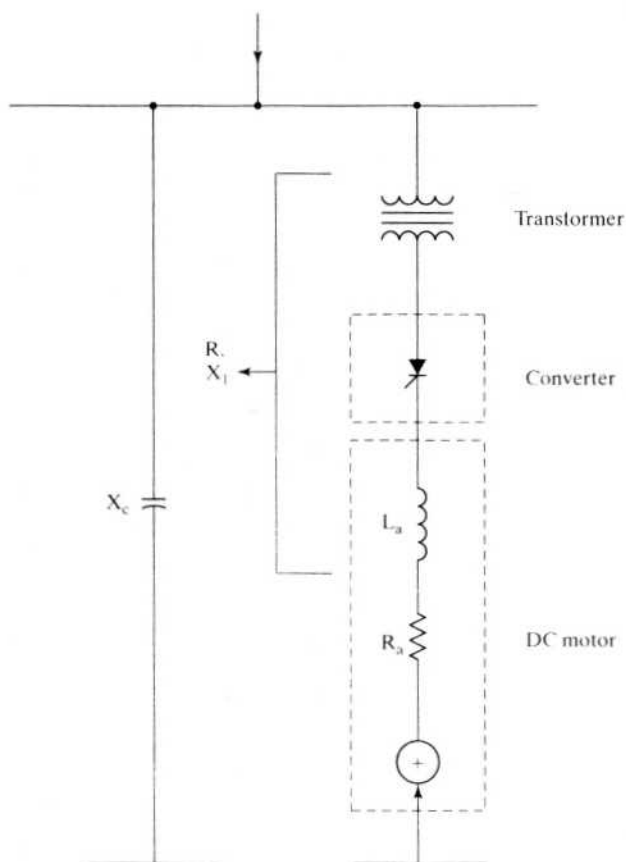


Figure 3.45 Schematic of the system network

That harmonic equivalent impedance does not change from the fundamental is to be noted in this network. Moreover, note the importance of the factor X_l/R in the calculation of the equivalent impedance of the network. By the same approach, the harmonic capacitor current I_{cn} and harmonic capacitor voltage V_{cn} for the n^{th} harmonic are derived in terms of the input current to the power system network, I_n , as

$$I_{cn} \approx j \left(\frac{X_{ln}}{R} \right) I_n \quad (3.159)$$

and

$$V_{cn} = \left(\frac{X_{ln}}{R} \right) X_{cn} I_n = \left(\frac{nX_l}{R} \right) \left(\frac{X_c}{n} \right) I_n \approx \left(\frac{X_l}{R} \right) X_c I_n \quad (3.160)$$

It is seen from the above that the ratio X_l/R is an important factor in such power-system calculations as the equivalent impedance, capacitor current, and capacitor voltage. Note that the reactance of the power-system network changes dynamically,

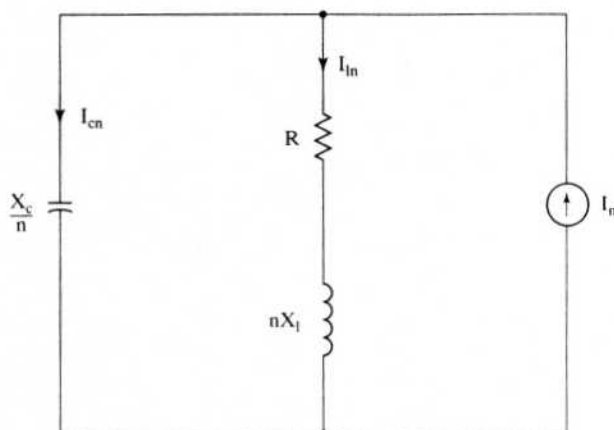


Figure 3.46 Harmonic equivalent of the system network

depending on the loads in the power system. The ratio X_l/R is calculated from the various impedances of the power-system network, such as source, transformer, transmission and distribution lines, and electrical motors and generators. The ratio X_l/R is calculated for the supply frequency. The harmonic currents generated by the phase-controlled converter, I_n , are calculated from the spectrum of the ac currents input to the phase-controlled converter as a function of the triggering angle and load conditions. Alternatively, the harmonic currents can be determined from the output harmonic voltages and harmonic impedances of the load. The harmonic currents are generated by the converter and hence are presented to the network as current sources, as is shown in Figure 3.46. The effect of resonance in this network is illustrated with the following example:

Line-to-line voltage, $V_l = 13.8$ kV

Capacitive reactance, $X_c = 8.96 \Omega$

X_l/R of the power system = 16

Fifth harmonic current, $I_5 = 60$ A

When $X_{15} = X_{c5}$,

$$I_{c5} = \left(1 + \frac{jX_{15}}{R}\right)(I_5) = \left(\frac{X_{15}}{R}\right)(I_5) = (5)\left(\frac{X_l}{R}\right)(I_5) = (5)(16)(60) = 4800 \text{ A}$$

The voltage across the capacitor bank is

$$V_c = \sqrt{V_{c1}^2 + V_{c5}^2} \quad (3.161)$$

where V_{c1} is the fundamental-phase voltage and V_{c5} is the phase voltage due to the fifth harmonic current, given by

$$V_{c1} = \frac{V_l}{\sqrt{3}} = \frac{13.8 \times 1000}{\sqrt{3}} = 7967 \text{ V}$$

$$V_{c5} = X_{c5} \times I_{c5} = \left(\frac{X_c}{5}\right)I_{c5} = \frac{8.96 \times 4800}{5} = 8601 \text{ V}$$

Hence,

$$V_c = \sqrt{7967^2 + 8601^2} = 11,724 \text{ V per phase}$$

which results in an overvoltage of 47%. This could result in exceeding the voltage and current rating of the capacitor bank. The higher currents could also exceed the circuit breaker rating. Both of these effects can lead to catastrophic failures in the power system, and such a case is cited in reference [15]. The resonance is indicated by the manifold increase in the harmonic currents flowing into the capacitor bank and the resulting overvoltage. Whether resonance might occur can be determined by finding the natural frequency of the power system and whether it coincides with any of the harmonic frequencies generated by the converter. Coincidence of the natural frequency with a harmonic frequency will result in the resonance. To evaluate such a condition, the natural frequency of the power system is determined as follows.

$$\frac{\text{short-circuit volt—ampere rating}}{\text{capacitor volt—ampere rating}} = \frac{\sqrt{3} \cdot V \cdot I_{sc}}{\sqrt{3} \cdot V \cdot I_c} \quad (3.162)$$

Neglecting the short-circuit resistance, and considering only the short-circuit reactance, X_{sc} , the ratio is given by

$$\frac{\text{short-circuit VA}}{\text{capacitor VA}} = \frac{V \cdot (V/X_{sc})}{V \cdot (V/X_c)} = \frac{X_c}{X_{sc}} = \frac{1}{(2\pi f_s)^2 \cdot L_{sc} \cdot C} \quad (3.163)$$

Let $f_o^2 = 1/((2\pi)^2 \cdot L_{sc} \cdot C)$, where L_{sc} is the short-circuit inductance and C is the capacitance of the capacitor bank, and substitute this relationship into the ratio of short-circuit-to-capacitor VA rating:

$$\frac{\text{short-circuit VA of the plant}}{\text{capacitor VA capacity}} = \frac{f_o^2}{f_s^2} = f_n^2 \quad (3.164)$$

where f_n is the natural frequency of the power-system plant expressed in multiples of supply frequency, f_s . The short-circuit rating of the plant is a variable depending on the topology of the network and system loads. To calculate the short-circuit rating of the plant, refer to any standard text on power-system analysis.

The harmonic resonance can be eliminated by attenuating the harmonic currents at their source of generation by installing LC filters selectively tuned to absorb the fifth and other harmonics of concern. Alternatively, inductors can be added in series with the capacitor banks to minimize the current inrush by changing their impedance. It is usual to install LC filters at the input to the converters. These LC filters should not affect the normal functioning of the rest of the power systems, as predetermined by the analysis of the network. This type of solution to the harmonic resonance is ideal for retrofit applications where the addition of new loads has changed the natural frequency. There are other solutions to the harmonics and their resonance. They are of two broad kinds.

Instead of using a six-pulse converter, it is possible to use a twelve-pulse converter, thereby eliminating the fifth and seventh harmonics completely. It is reminded that the most dominant harmonic creating resonance-related problem in the installations is usually the fifth, very rarely the seventh. By eliminating these harmonics, the basic problem of resonance is tackled. The twelve-pulse converter is

made up of two six-pulse converters in parallel. The converter sets can be fed through an isolation transformer with two secondaries, one of which is delta-connected, the other star-connected, to provide the displacement in the voltages and hence the steps in the input current drawn from the ac supply. This form of solution to the harmonics is passive and uses a time-honored reliable converter topology with line commutation. Even though the cost of the system will go up (two converter sets, isolation transformer with two secondaries, etc.), there will be considerable reduction in the filter size, and this converter configuration is ideal for high power. The twelve-pulse controlled converter is considered in the following subsection in some detail.

The other kind of harmonic control involves the switching control of the converters. This requires either forced commutation of the SCR switches or self-commutation by using transistors, GTOs, IGBTs, MOSFETs, etc. By a switching of the supply input in a selective manner, the dominant harmonics are selectively eliminated. This is an active solution, but it will involve additional cost compared to the line-commutated converters. Perhaps with time the cost of the self-commutating switches will come down further, to make this option economically attractive for future installations.

3.12.2 Twelve-Pulse Converter for DC Motor Drives

Two six-pulse phase-controlled converters deriving their ac inputs from a set of star (wye) delta-connected secondaries whose primary is delta-connected constitute a twelve-pulse converter. The power schematic is shown in Figure 3.47. The output of the star-connected secondary voltages lags the delta-connected secondary voltages by 30 degrees. The converter outputs are connected through an interphase reactor. This reactor limits the circulating current between the bridge converters and further serves the purpose of paralleling the two outputs of different magnitude provided by the bridge converters. The output voltage is the average of the two bridge converters' outputs.

The control of the bridges is very similar to the control of a six-pulse phase-controlled converter. The operation of the twelve-pulse converter is explained by using Figure 3.48. When v_{a1b1} is positive and 60 degrees from its positive-zero crossing, devices 11 and 12 can be turned on. With a triggering delay angle α , the output is v_1 . This output, v_1 , is 60 electrical degrees in duration following the line voltage, v_{a1b1} , and, hence, has six pulses per cycle. The output v_2 is very similar to v_1 but is phase-displaced 30 electrical degrees by the phase shift of their ac input voltages. The interphase reactor averages these two output voltages to provide the resultant load voltage v_a , indicated by dashed lines in Figure 3.48. The load voltage contains a twelfth harmonic. The input ac current contains zero 5th and 7th harmonic, as there is no reflected sixth-harmonic current in the load. Hence the minimum higher harmonic present in the ac input is the eleventh. The average output voltage of this twelve-pulse converter for continuous load current is

$$v_a = \frac{v_1 + v_2}{2} \quad (3.165)$$

where the average bridge converter outputs are

$$v_1 = v_2 = 1.35V \cos \alpha \quad (3.166)$$

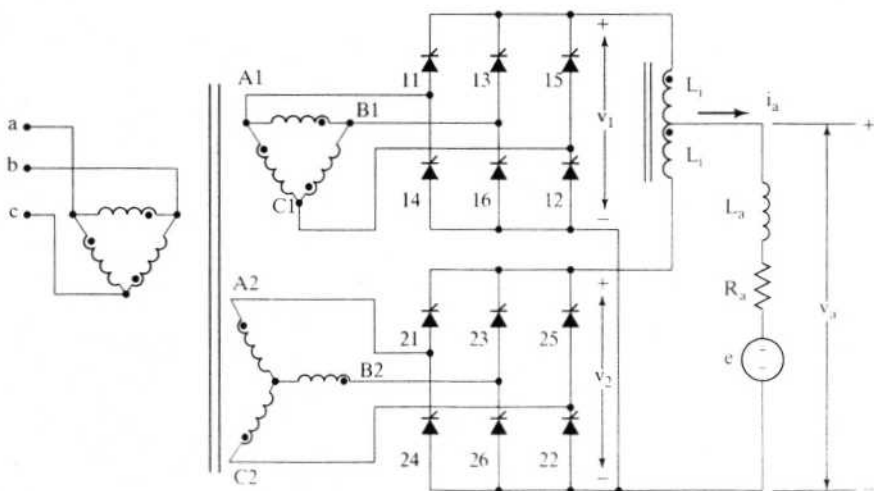


Figure 3.47 Twelve-pulse converter for dc drive

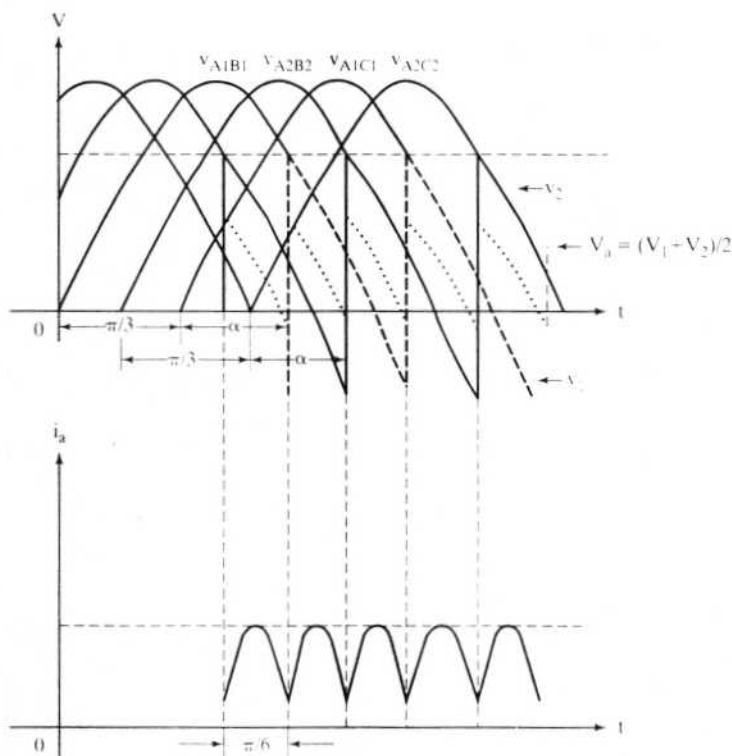


Figure 3.48 Voltage and current waveforms of the 12-pulse converter

Hence,

$$v_a = 1.35V \cos \alpha \tag{3.167}$$

The twelve-pulse converter eliminates the lower harmonics ($< \text{eleventh}$) but does not affect the power factor. This implies that, at low-speed operation of the motor requiring low dc output voltage, the triggering angle α will be close to 90 degrees. This will result in low power factor, and it will be dependent on the motor speed, very much as is the six-pulse converter-controlled dc motor. The improvement in power factor requires additional switching of the bridge converter to control the reactive power, and that is described in the next section.

3.12.3 Selective Harmonic Elimination and Power Factor Improvement by Switching

Increasing the number of switchings in a cycle eliminates the harmonics lower than the switching frequency. For instance, a six-pulse converter has no harmonics below 6 at the output and 5 at the input. Simultaneously, the phase of the input current with respect to its phase voltage can be controlled, resulting in power-factor control and obtaining a desirable operation near or at unity power factor. Such a converter circuit has to have self-commutating switches, i.e., switches capable of turning off regardless of the polarity of the voltage and current. Consider a converter with GTOs, shown in Figure 3.49 with input filter network. The filter contains an inductor L_f and a capacitor C_f per phase. The source has an inductance L_s per phase. Since turn-off is achieved by gate control of the GTO, the switching constraint of the six-pulse SCR converter is removed in this arrangement. The switches can be gated if the anodes are positive with respect to the cathodes over and above their conduction voltage drops. Suppose T_1 and T_2 are conducting and are to be turned off. If the armature current is continuous, a path for its continued flow needs to be ensured. Otherwise, a large voltage due to the inductive drop will arise and appear across the switches, resulting in their destruction. It is averted by gating T_3 and T_6 and shorting

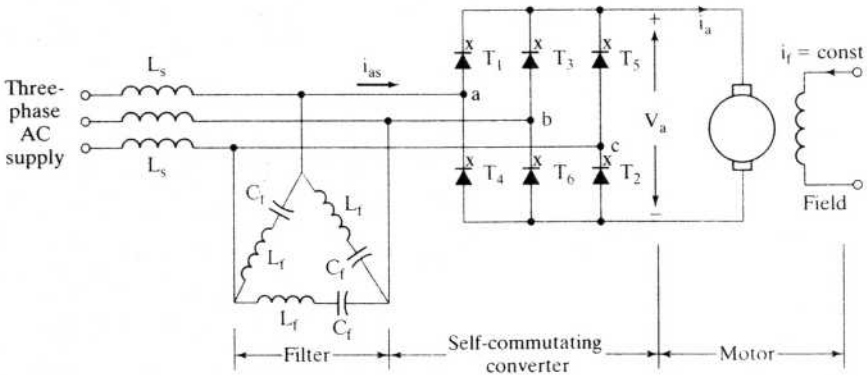


Figure 3.49 Self-commutating converter-controlled dc motor drive

the armature. Due to the delays in turning off T_1 and turning on T_3 , an instant may occur with no path for the current, resulting in a large voltage across the devices. That is avoided by overlapping the control signals of T_1 and T_3 . As for the ac side, the filter provides a path for the converter input current during device turn-off. During turn-on of a pair of switches, the voltage across the filter will appear across the load, resulting in a large inrush current. This may very well exceed the critical rate of rise of the current of the switches. To prevent such an effect, the filter is provided with the inductor L_f .

The output voltage is controlled by the conduction duration of the power devices. This is in contrast to phase control, in which the output voltage is varied by moving the 60-degree segment of voltage to give the desired average output voltage. Such a movement of the voltage segment results in a movement of current and hence in the phase displacement of the current and a poor power factor. In the present arrangement, the voltage and current are not moved, resulting in zero phase shift of the current. This is illustrated with Figure 3.50. The control of the voltage starts at 60 degrees from the positive-zero crossing of the line voltages, as in the case of the phase control. Each GTO is used for 120 degrees, during which the device *on* and *off* times are varied to provide the desired average voltage to the machine armature. The dc output voltage contains approximately the first harmonic at switching frequency, and the ripple current is minimized by appropriately choosing the maximum available switching frequency consistent with the switch characteristics and loss. The reduction of harmonic armature currents means that a dc reactor might not be necessary in this converter arrangement, hence, none is shown in Figure 3.49. The phase current, i_{as} , consists of the on-time currents of the armature only.

The control of output voltage requires consideration of the turn-on and turn-off times of the power switches. There is a minimum duration of conduction required for the switch, below which the switching losses would dominate the quantum of power transfer from the source to load or vice versa. Hence, low-speed operation requiring low voltages will have problems with pulse-width control. In that region, then, it is necessary to adopt phase control. This comes at the expense of low power factor and high harmonics. The encouraging factor in this mode of operation is that it is required only for less than 10% of the speed range in many applications, beyond which pulse-width control can be resorted to. The control is identical for regeneration mode for both the pulse-width and phase control of this converter.

Care must be exercised in the selection of filter component ratings and snubber ratings for the system, to avoid a resonance due to the coincidence of the switching frequency with the natural frequency of the circuit or the resonant frequency of the filter.

These converter configurations are in practice at 600-kVA ratings for some elevator applications. The ratings of these converters will increase in the future, as the harmonic reduction and power-factor improvement required by the utilities for energy savings become a key economic factor in the day-to-day operation of industries. To reduce the cost of these converter arrangements, the lower half of the bridge can have SCR switches; only the upper half of the bridge need have self-commutating switches.

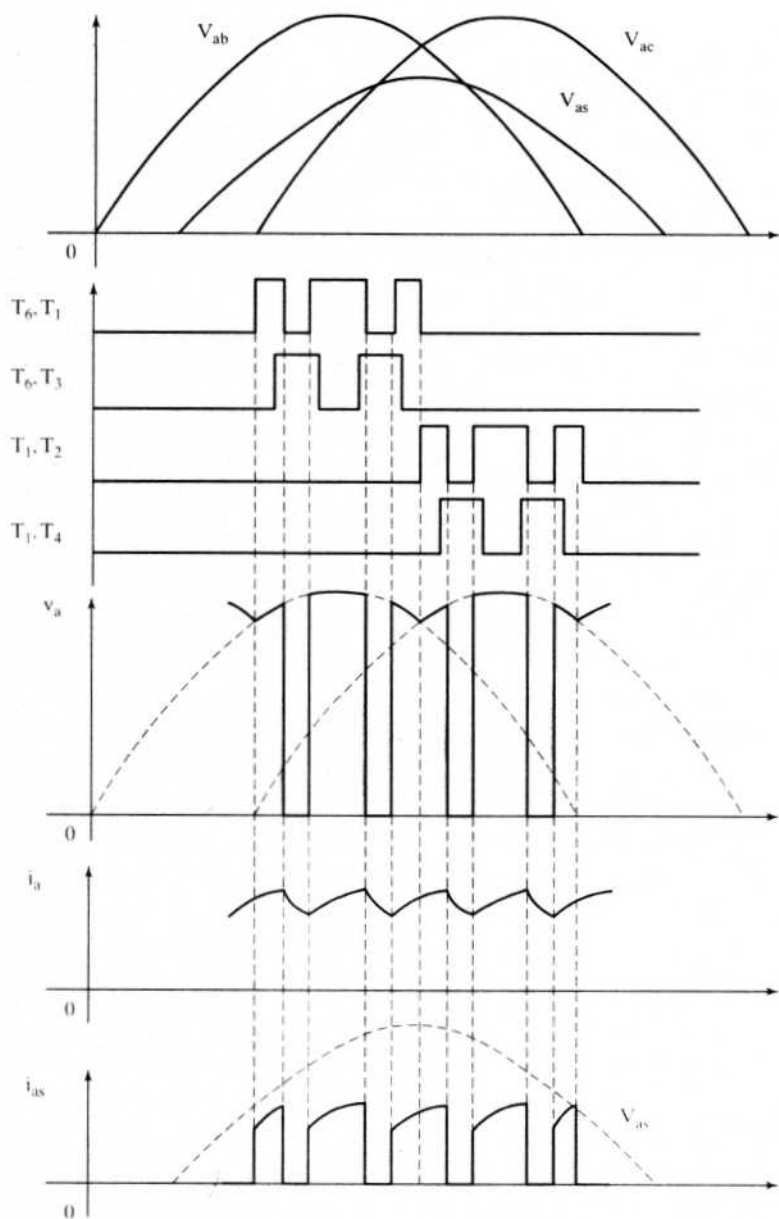


Figure 3.50 Waveforms of the self-commutating converter-controlled dc motor drive

3.13 SIXTH-HARMONIC TORQUE

The previous section dealt with the input-side harmonics of the phase-controlled converter and the associated problems. This section presents the effect of the harmonics on the output side of the six-pulse phase-controlled converter, i.e., the input harmonics to the dc motor. The dominant harmonic in the six-pulse converter is the sixth harmonic, and only that one is considered here. Compared to the effects of the sixth harmonic, the effects of other harmonics are smaller, because the harmonic impedances are very high for higher-order harmonics and thus result in the attenuation of harmonic currents. The approach for calculating the effects of the other harmonics can follow along the same lines as the calculation for the effects of sixth harmonic developed in the following. The effect of the dominant harmonic is to produce a torque whose average is zero and which is therefore useless from the point of view of driving a load. It also creates losses in the form of heating by increasing the rms value of the input armature current. The harmonic torque can contribute to speed ripples at low speed if the mechanical impedance is not adequate to filter it. It can also excite some of the resonant frequencies of the load and disable the smooth functioning of the motor-drive system. It is essential to evaluate the magnitude of the harmonic torque and its effect on the armature heating for application considerations. For both the continuous and discontinuous current-conduction modes, the relevant equations are derived; some are used from the previous section on steady-state analysis.

3.13.1 Continuous Current Conduction Mode

The armature voltage and current waveforms corresponding to Figure 3.22 (given in section 3.4.2) are considered for harmonic analysis. The armature voltage can be resolved into Fourier series as

$$v_a = \sum_{n=1}^{\infty} (a_n \sin n\theta + b_n \cos n\theta) + a_{01} \quad (3.168)$$

where

$$a_{01} = V_a = 1.35V \cos \alpha \quad (3.169)$$

$$a_n = \frac{3V_m}{\pi} \left[\frac{\sin \frac{n-1}{2}\theta}{n-1} - \frac{\sin \frac{n+1}{2}\theta}{n+1} \right] \frac{2\pi}{3} - \alpha \quad (3.170)$$

$$b_n = \frac{3V_m}{\pi} \left[\frac{\cos \frac{n-1}{2}\theta}{n-1} - \frac{\cos \frac{n+1}{2}\theta}{n+1} \right] \frac{2\pi}{3} - \alpha \quad (3.171)$$

where n is the harmonic number and from which the sixth harmonic components are

$$a_6 = \frac{3\sqrt{2}V}{\pi} \left[\frac{\sin 7\alpha}{7} - \frac{\sin 5\alpha}{5} \right] \quad (3.172)$$

$$b_6 = \frac{3\sqrt{2}V}{\pi} \left[\frac{\cos 7\alpha}{7} - \frac{\cos 5\alpha}{5} \right] \quad (3.173)$$

The peak value of the sixth-harmonic armature voltage is

$$V_{6p} = \sqrt{a_6^2 + b_6^2} \quad (3.174)$$

which, upon substitution of (3.172) and (3.173), yields

$$V_{6p} = \frac{3\sqrt{2}V}{\pi} \cdot \frac{1}{35} \sqrt{74 - 70 \cos 2\alpha} \quad (3.175)$$

This peak sixth-harmonic armature voltage is maximum at a triggering angle of 90 degrees and is given as

$$V_{6p} = 0.463 V \quad (3.176)$$

The sixth-harmonic current and electromagnetic torque are derived as

$$I_6 = \frac{V_{6p}}{Z_6} \quad (3.177)$$

$$T_{e6} = K_b I_6 \quad (3.178)$$

where the sixth-harmonic impedance is given by

$$Z_6 = R_a + j6\omega_s L_a \cong j6\omega_s = 6X_a \quad (3.179)$$

The peak sixth-harmonic torque expressed in p.u. is

$$\begin{aligned} T_{e6n} = \frac{T_{e6}}{T_{er}} &= \frac{K_b \frac{3\sqrt{2}V}{\pi} \cdot \frac{1}{35} \sqrt{74 - 70 \cos 2\alpha}}{6X_a K_b I_{ar}} = 0.00643 \frac{V/V_r}{X_a I_{ar}/V_r} \sqrt{74 - 70 \cos 2\alpha} \\ &= 0.00643 \frac{V_n}{X_{an}} \sqrt{74 - 70 \cos 2\alpha} \end{aligned} \quad (3.180)$$

where V_n is the normalized voltage and X_{an} is the normalized armature reactance given by

$$V_n = \frac{V}{V_r} \text{ p.u.} \quad (3.181)$$

$$X_{an} = \frac{I_{ar} X_a}{V_r} \text{ p.u.} \quad (3.182)$$

Note that the sixth-harmonic torque is a function of the triggering angle, which would change with the induced emf and armature current. Hence, the triggering angle corresponding to the induced emf and armature current has to be evaluated to find the sixth-harmonic torque. For a large dc machine with a base voltage of 500 V, base current of 2650 A, 0.027 p.u. resistance, 0.2 p.u. inductance, emf constant of 8V/rad/sec. and input rms line-to-line voltage of 560 V, the peak sixth harmonic is computed; it is shown in Figure 3.51. The torque pulsations at low speeds are higher in magnitude than those at high speeds, because the triggering angle will become higher with decreasing speeds, thus contributing to increasing sixth-harmonic voltage. The fact that the torque pulsation can be as high as 0.42 p.u. at zero speed can make it unsuitable for high-precision applications. If need be, the pulsations can be

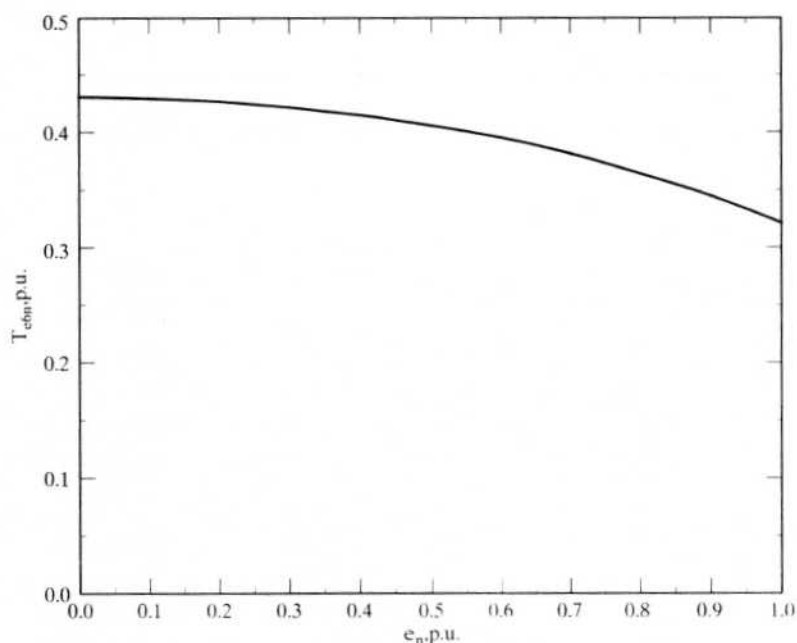


Figure 3.51 Normalized peak sixth-harmonic torque vs. induced emf

reduced by increasing the harmonic impedance with an external reactor. The other solution for decreasing the harmonic voltage is dealt with in the next chapter: using a chopper control.

Calculation of armature resistive loss: The rms armature current is found from

$$I_{\text{rms}} = \sqrt{I_{\text{av}}^2 + I_{6p}^2/2} \quad (3.183)$$

where the average dc current is obtained from

$$I_{\text{av}} = \frac{1.35V \cos \alpha - E}{R_a} \quad (3.184)$$

The ratio of armature resistive loss for phase controlled input to a pure dc input is

$$\frac{I_{\text{av}}^2 + \frac{I_6^2}{2}}{I_{\text{av}}^2} = 1 + \frac{1}{2} \left(\frac{I_{6p}}{I_{\text{av}}} \right)^2 \quad (3.185)$$

where

$$\frac{I_{6p}}{I_{\text{av}}} = \frac{V_6/6X_a}{(1.35V \cos \alpha - E)/R_a} \quad (3.186)$$

By substituting for V_6 , the ratio of currents is

$$\frac{I_{6p}}{I_{av}} = \frac{0.00643}{\tan \beta} \cdot \frac{\sqrt{74 - 70 \cos 2\alpha}}{1.35 \cos \alpha - E/V} \quad (3.187)$$

The increase in armature resistive loss due to the sixth-harmonic component is

$$\frac{1}{2} \left(\frac{I_{6p}}{I_{av}} \right)^2 = 0.2067 \times 10^{-4} \left[\frac{74 - 70 \cos 2\alpha}{(1.35 \cos \alpha - E/V)^2 \cdot \tan^2 \beta} \right] \quad (3.188)$$

This is difficult to interpret when the average current becomes zero; hence, an alternative formulation is given below. The normalized armature resistive loss is

$$\begin{aligned} P_{cn} &= \frac{P_c}{P_{cr}} = \frac{I_{av}^2 + I_{6p}^2/2}{I_{ar}^2} = \frac{\left(\frac{1.35V \cos \alpha - E}{R_a} \right)^2 + \frac{0.00643^2}{2} \cdot \frac{V^2}{X_a^2} (74 - 70 \cos 2\alpha)}{(V_b/Z_b)^2} \\ &= \left(\frac{1.35V_n \cos \alpha - E/V_b}{R_{an}} \right)^2 + \frac{0.2067 \times 10^{-4} V_n^2}{X_{an}^2} (74 - 70 \cos 2\alpha) \end{aligned} \quad (3.189)$$

and this equation is valid for all

$$a \leq \cos^{-1} \left(\frac{E}{1.35V} \right), \text{ rad} \quad (3.190)$$

If $V_r = V$, then $V_n = 1$, and hence the normalized armature resistive loss is given by

$$P_{cn} = \left(\frac{1.35V_n \cos \alpha - e_n}{R_{an}} \right)^2 + \frac{0.2067 \times 10^{-4}}{X_{an}^2} (74 - 70 \cos 2\alpha) \cdot \text{p.u.} \quad (3.191)$$

where the normalized induced emf and normalized resistance and reactance are

$$e_n = \frac{E}{V} \cdot \text{p.u.} \quad (3.192)$$

$$R_{an} = R_a/Z_b \cdot \text{p.u.} \quad (3.193)$$

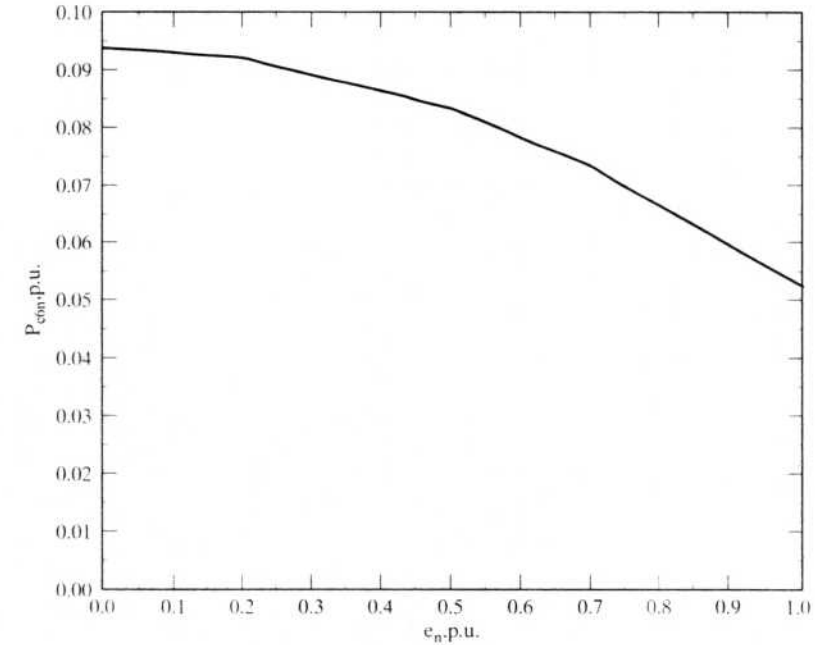
$$X_{an} = X_a/Z_b \cdot \text{p.u.} \quad (3.194)$$

For the same machine given in the previous T_{con} calculations, the normalized armature resistive loss against e_n is calculated with the above procedure, as shown in Figure 3.52. The average armature current is assumed to be 1 p.u. The copper losses have increased by approximately 9% at zero speed, and at 1 p.u. speed they have increased only by 5%. For continued operation, this increase in losses has to be compensated for by derating the machine, and note this is not negligible.

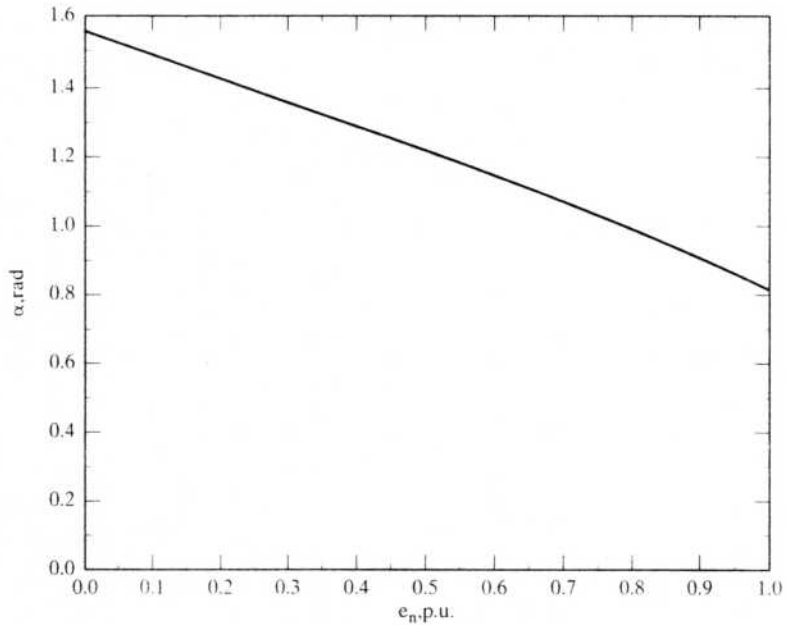
3.13.2 Discontinuous Current-Conduction Mode

The discontinuous current-conduction mode has been dealt with in the steady-state analysis section. The normalized average and sixth-harmonic voltages are found, from which the respective currents are derived as

$$I_{avn} = (V_{an} - E_n) \cdot \frac{\theta_s}{R_{an} \cdot \pi/3} \quad (3.195)$$



(i) Armature resistive losses due to sixth-harmonic current



(ii) Triggering angle

Figure 3.52 Peak sixth-harmonic armature resistive losses and triggering angle in normalized units, as a function of normalized induced emf for continuous conduction

where the normalized dc input voltage is

$$V_{an} = \sqrt{2}V_n \sqrt{2 - 2 \cos \theta_x} \cdot \frac{\cos(\pi/3 + \alpha - \theta_1)}{\theta_x} \quad (3.196)$$

$$\theta_1 = \tan^{-1} \left(\frac{\sin \theta_x}{1 - \cos \theta_x} \right) \quad (3.197)$$

Note that the average normalized motor input voltage is for the conduction duration, since only it contributes to the current. The current is averaged over the period of $\pi/3$ instead of θ_x . The sixth-harmonic current has a peak given by

$$I_{6pn} = \frac{V_n}{X_{an}} \cdot \left[\frac{1}{\pi\sqrt{2}} (a_1 + a_2) - 2\sqrt{a_1 \cdot a_2} \cos(2\alpha + 2 \cdot \pi/3 + \theta_3 - \theta_2) \right] \quad (3.198)$$

where

$$a_1 = \frac{2 - 2 \cos 7\theta_x}{49} \quad (3.199)$$

$$a_2 = \frac{2 - 2 \cos 5\theta_x}{25} \quad (3.200)$$

$$\theta_2 = \tan^{-1} \left(\frac{\sin 5\theta_x}{\cos 5\theta_x - 1} \right) \quad (3.201)$$

$$\theta_3 = \tan^{-1} \left(\frac{\sin 7\theta_x}{\cos 7\theta_x - 1} \right) \quad (3.202)$$

The triggering angle should be greater than its critical value, derived earlier, to ensure that the operating point is in the discontinuous current-conduction mode. The peak normalized sixth-harmonic torque is equal to the normalized peak sixth-harmonic current, and the resistive loss due to it in the armature is given in normalized units as

$$P_{6pn} = \frac{I_{6pn}^2}{2}, \text{ p.u.} \quad (3.203)$$

The algorithm for the computation of the sixth-harmonic torque is as follows: (i) Find the impedance angle β and the critical triggering angle α_c . (ii) Choose an α greater than α_c that will give the desired average armature current for the assumed normalized induced emf. (iii) Find the conduction angle, θ_x , using equation (3.60), by Newton–Raphson technique. (iv) The sixth-harmonic current and the corresponding armature losses are evaluated from the expression derived above. For the same machine parameters as in the continuous-conduction mode and an average armature current of 1.15 p.u., the triggering angle (α) and conduction angle (γ) vs. normalized induced emf are shown in Figure 3.53. The peak normalized sixth-harmonic torque and its corresponding loss vs. the normalized induced emf are shown in Figure 3.54. The maximum harmonic torque occurs at

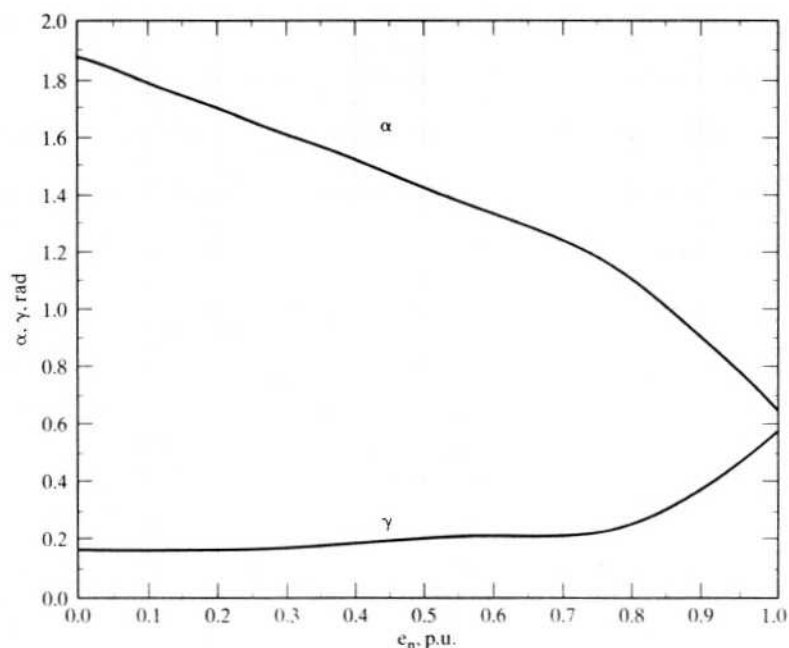


Figure 3.53 Triggering angle and conduction angle vs. induced emf for discontinuous conduction

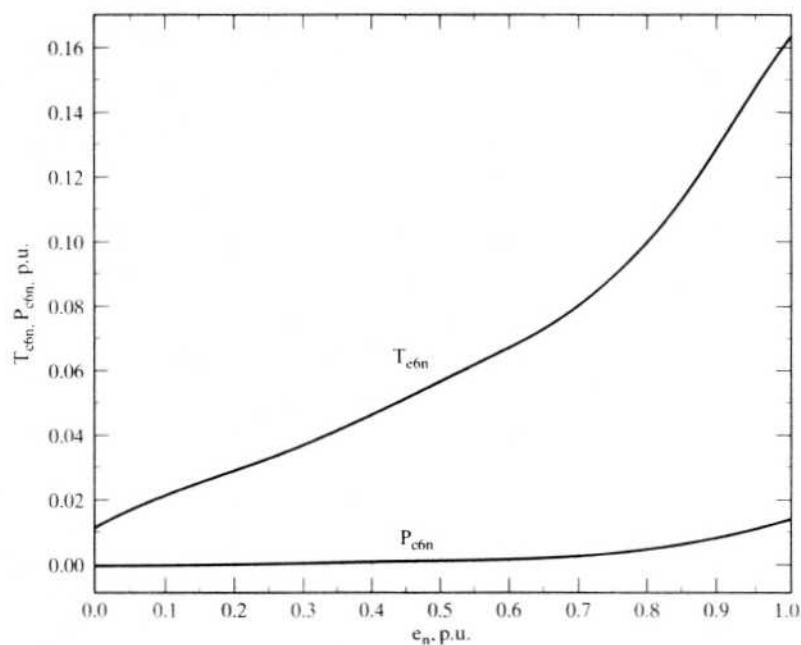


Figure 3.54 Normalized peak sixth-harmonic torque and armature resistive loss vs. induced emf in discontinuous conduction

1 p.u. induced emf or at 1 p.u. speed and has a minimum value at zero speed. This is contrary to the continuous current-conduction mode, where the maximum sixth-harmonic torque occurs at zero speed and the minimum occurs at 1 p.u. speed. The maximum sixth-harmonic torque is approximately 0.16 p.u. for the example considered; that is very high for many high-performance applications. The harmonic currents and torques are lower in the discontinuous current-conduction mode compared to the continuous current-conduction mode, for this machine.

3.14 APPLICATION CONSIDERATIONS

The torque–speed characteristics in steady-state and intermittent operation provide the power and speed ratings of the dc motor. Torque–speed characteristics translate into voltage and current ratings of the motor and converter. The controller design coordinates the steady-state with the dynamic requirements of the motor drive. There are a number of factors to be considered in the selection of the motor, converter, and controller. Some of these factors are listed in the following, from reference [19]:

1. Motor

- Power, hp.
- Service factor
- Armature and field voltage
- Armature and field current
- Efficiency
- Preferred direction of rotation
- Rated speed
- Maximum speed
- Motor friction and inertia
- Load friction and inertia
- Type of enclosure
- Class of insulation
- Cooling arrangements
- Safe temperature rise
- Tachometer characteristics
- Encoder details
- Brush resistance, life span, and maintenance
- Mounting details
- Gears for speed reduction
- Protection features

2. Converter

- Input ac voltage, output dc voltage
- RMS current rating
- Maximum current rating (short-time rating)
- Efficiency vs. load
- Cooling
- Safe thermal limits

- Protection features
 - Overvoltage
 - Overcurrent
 - Contactors
 - Enclosure
3. Controller
- Features such as regeneration
 - Reversal of speed
 - Steady-state accuracy or error
 - Dynamic responses (overshoot, bandwidth, rise time, etc.)
 - Flexibility to change controller gains
 - Metering and data logging
 - Protection coordination of the motor and converter
 - Interface capability to other processes
 - Loss of field supply, mains supply, etc.

The motor might have to be derated to account for the additional losses produced by the harmonic currents. This derating has to be a function of load and of the type of converter used.

3.15 APPLICATIONS

Because the phase-controlled dc motor drive is one of the earliest and most reliable variable-speed drives, it has found a large number of industrial applications. An example of its application is described here to impart a sense of reality to the theory and analysis described above.

A schematic for a flying shear is shown in Figure 3.55. Flying shears are used in the metal-rolling industry for cutting materials to desired lengths to suit customer needs. A flying shear consists of two blades rotating in opposite directions from their initial positions of c and c' at a velocity of V_b m/s. The material to be cut moves in at a velocity of V_m m/s. The blades are driven by the shafts of two separately-excited dc motor drives connected in series and controlled from a dual set of

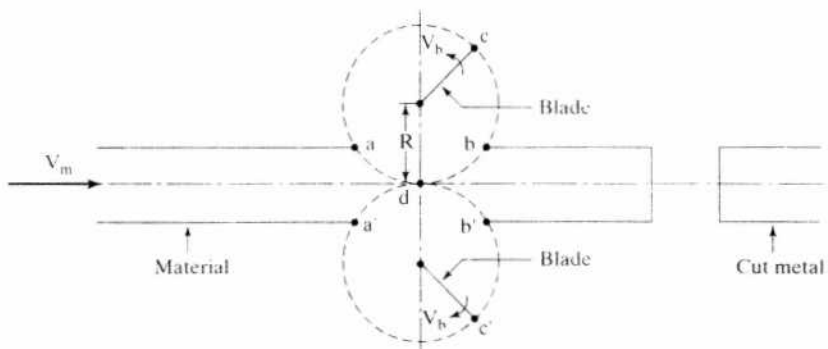


Figure 3.55 Flying-shear system schematic

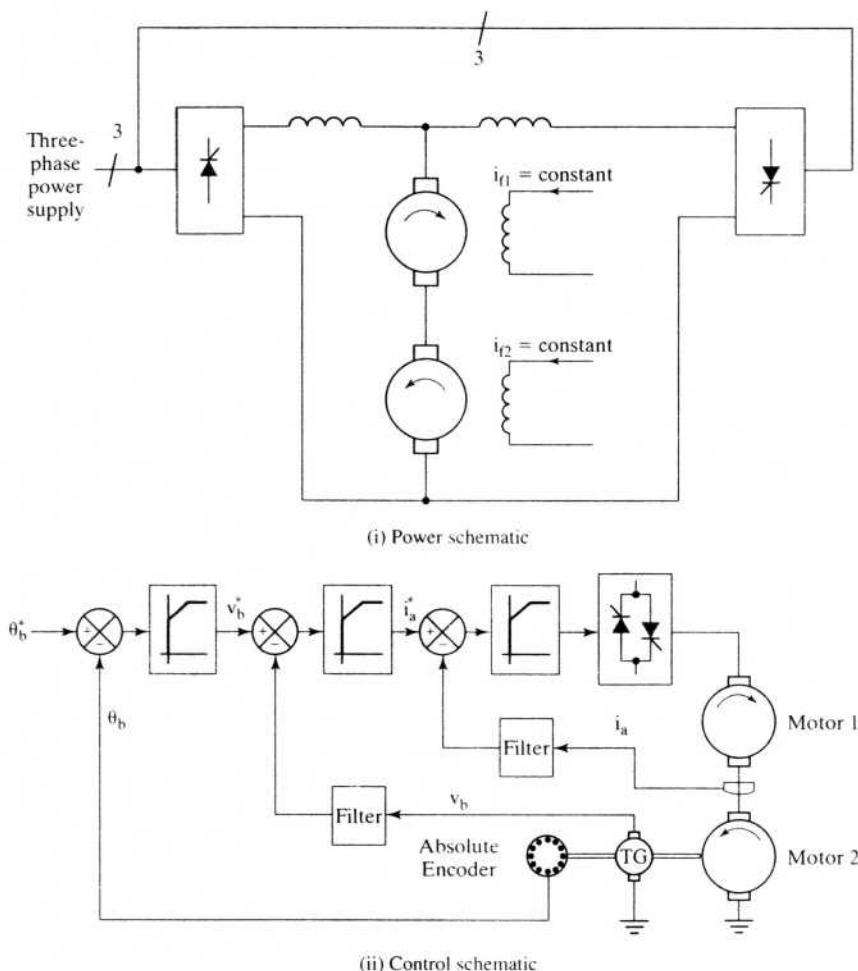


Figure 3.56 Flying-shear control system

three-phase controlled-rectifier bridges, as shown in the power schematic of Figure 3.56. When the material is to be cut, the blades are accelerated until they come to positions a and a' . The acceleration is made to be a function of the velocity of the material inflow, V_m , of the acceleration of the blades, V_b , and of the positions of the blades and the material. A slight deceleration is initiated at this point as the shearing starts and the blades reach point d . At position d , the front end of the material has been cut off. In order not to be in the way obstructing the flow of the material, the blades are accelerated till they reach points b and b' , and then they are decelerated to slide into their original positions c and c' . The drive systems controlling the blades have to be highly responsive and are position-controlled drive systems. The key variables of the system are shown in Figure 3.57 for a cycle of operation. The blade velocity exceeds 20 m/s nominally for a short period, and at the time of the metal cutting it equals the velocity of the material. The armature-current commands for the dc

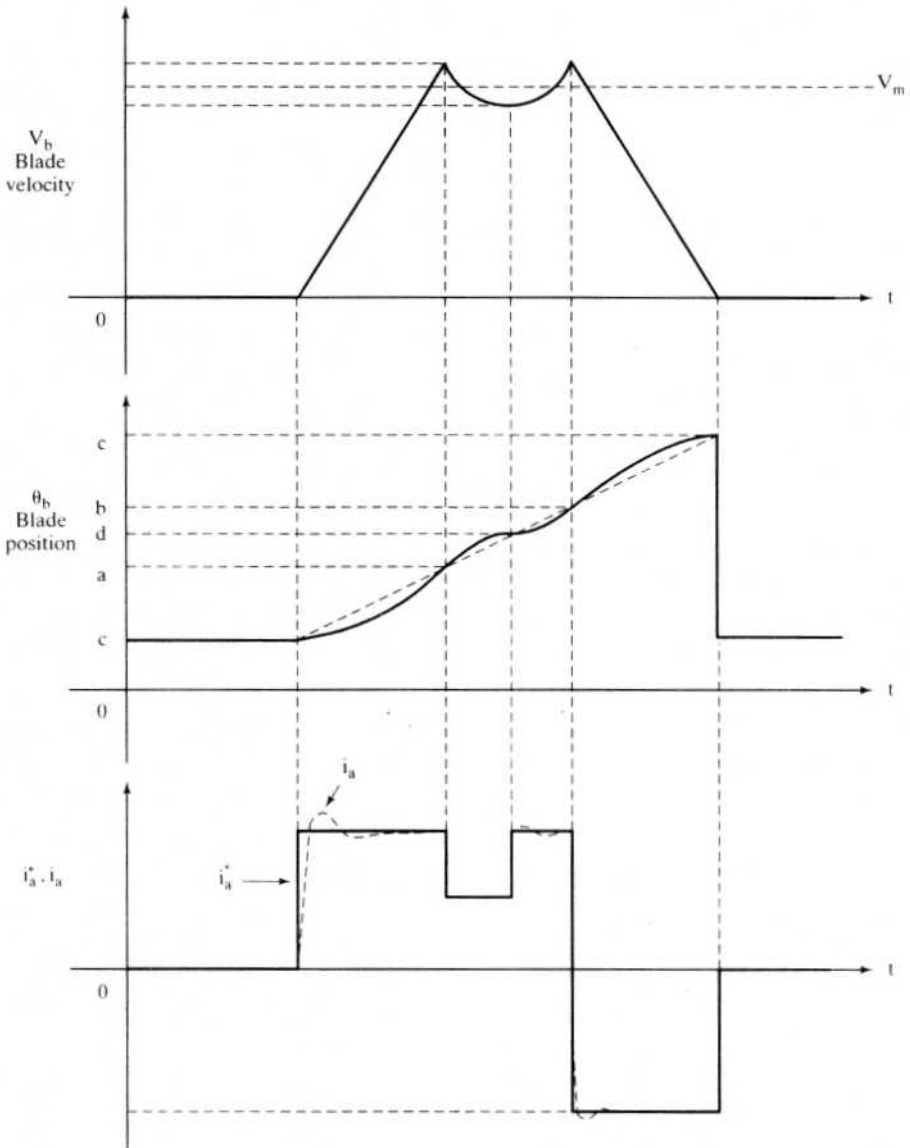


Figure 3.57 Flying-shear system response

motors are at positive and negative maxima during acceleration and deceleration, respectively. The actual motor currents have to follow their commands closely for precise operation of the flying shear. This requires a fast-acting current loop, i.e., a very high current-loop bandwidth for the system. For example, the flying-shear drive system might consist of two 1250-k W motors of 600 V. The rated current of this system is approximately 2200 A. Current reversals and ramping at this magnitude within a few milliseconds, say 50 to 100 ms, are desirable for a fast-acting flying-shear system.

3.16 PARAMETER SENSITIVITY

A change in temperature increases the armature resistance of the dc machine. In the closed-loop drive, this will not have an effect on such output variables as torque and speed. It can increase the armature resistive loss; for example, a temperature rise of 150°C varies the resistance and loss by more than 70% from its nominal value at ambient temperature. Therefore, reconsideration of the thermal limits of the machine and of derating is essential.

In an open-loop motor drive, the changes in armature resistance will decrease the rotor speed. Assuming that the load torque is a constant amounts to a constant armature current; hence, the induced emf decreases for a given input voltage, because the resistance drop has increased.

Decreasing induced emf means a reduction in rotor speed. Note that the ambient conditions and changing conditions are denoted by the subscripts 0 and c, respectively.

$$\omega_{m0} = \frac{e_0}{K_b} = \frac{V_a - i_{a0}R_{a0}}{K_b} \quad (3.204)$$

$$\omega_{mc} = \frac{e_c}{K_b} = \frac{V_a - i_{a0}R_{ac}}{K_b} \quad (3.205)$$

The change in speed due to parameter sensitivity is

$$\frac{\omega_{mc}}{\omega_{m0}} = \frac{V_a - i_{a0}R_{ac}}{V_a - i_{a0}R_{a0}} \quad (3.206)$$

The changing variables can be written as a sum of ambient or normal values and the incremental changes:

$$\omega_{mc} = \omega_{m0} + \delta\omega_m \quad (3.207)$$

$$R_{ac} = R_a + \delta R_a \quad (3.208)$$

When $\delta\omega_m$ is the change in speed due to δR_a change in armature resistance, then by substituting equation (3.207) and (3.208) into (3.206) we get

$$1 + \frac{\delta\omega_m}{\omega_{m0}} = 1 - \frac{i_{a0}\delta R_a}{V_a - i_{a0}R_a} \quad (3.209)$$

from which the incremental change in speed is expressed as

$$\frac{\delta\omega_m}{\omega_{m0}} = - \frac{i_{a0}\delta R_a}{e_0} = - \frac{i_{a0}\delta R_a}{K_b\omega_{m0}} \quad (3.210)$$

Hence,

$$\delta\omega_m = - \left[\frac{i_{a0}}{K_b} \right] \delta R_a \quad (3.211)$$

Likewise, the effect due to saturation can be computed. In the case of saturation, the field flux will change and hence so will the emf constant K_b .

The dynamic performance is affected along with the changing armature resistance and saturation level. That is bound to affect the transient response if the controllers are fixed.

3.17 RESEARCH STATUS

Current research is focused mainly on the following aspects of the dc motor drive:

- (i) Adaptive controllers
- (ii) Improvement of power factor and reduction of harmonics

Adaptive controllers have become necessary to meet changing load conditions without affecting dynamic performance. This is evident, for example, in the variation of inertia as a robot's arm is gradually moved. A number of processes need to avoid torsional resonance, and they can be controlled by changing either the controller configuration or the controller gains. New control algorithms using fuzzy, neural, and other nonlinear techniques are on the rise; most of them are implemented with microprocessors.

With devices such as GTOs, transistors, and MOSFETs, it is possible to resort to pulse-width modulation of the input ac voltage to eliminate the harmonics selectively. That would keep the input current more nearly sinusoidal and the power factor close to unity. Even though such switching techniques increase the complexity of control and the switching losses, they are worth investigating with the devices and controllers now becoming cost-effective.

3.18 SUGGESTED READINGS

1. David L. Duff and Allan Ludbrook, "Reversing thyristor armature dual converter with logic crossover control," *IEEE Trans. Industry and General Application*, vol. IGA-1, no. 3, pp. 216–222, 1965.
2. T. Krishnan and B. Ramaswami, "A fast-response dc motor speed control system," *IEEE Trans. Industry Applications*, vol. IA-10, no. 5, pp. 643–651, Sept./Oct. 1974.
3. T. Krishnan and B. Ramaswami, "Speed control of dc motor using thyristor dual converter," *IEEE Trans. Industrial Electronics and Control Instrumentation*, vol. IECI-23, no. 4, pp. 391–399, Nov. 1976.
4. Geza Joos and T.H. Barton, "Four-quadrant dc variable speed drives—design considerations," *Proc. IEEE*, vol. 63, no. 12, pp. 1660–1668, Dec. 1975.
5. J. K. Haggerty, J. T. Maynard, and L. A. Koenig, "Application factors for thyristor converter dc motor drives," *IEEE Trans. Industry General Applications*, vol. IGA-7, pp. 718–728, Nov./Dec. 1971.
6. J. P. Sucena-Paiva, R. Hernandez, and L.L. Frerics, "Stability study of controlled rectifiers using a new discrete model," *Proc. Institute of Electrical Engineers*, London, vol. 119, pp. 1285–1293, Sept. 1972.
7. C. E. Robinson, "Redesign of dc motors for applications with thyristor power supplies," *IEEE Trans. Industry and General Applications*, vol. IGA-4, pp. 508–514, Sept./Oct. 1968.

8. Leland A. Schlabach, "Conduction limits of a three-phase controlled converter in inversion," *IEEE Trans. Industry Applications*, vol. IA-22, no. 2, pp. 298–303, March/April 1986.
9. B. R. Pelly, *Thyristor Phase-Controlled Converters and Cycloconverter*, New York, Wiley, 1971.
10. S. B. Dewan and W. G. Dunford, "Improved power factor operation of a three-phase rectifier bridge through modified gating," *Conference Record, IEEE-IAS Annual Meeting*, pp. 830–837, Oct. 1980.
11. B. Ilango, R. Krishnan, R. Subramanian, and R. Sadasivam, "Firing circuit for three-phase thyristor bridge rectifier," *IEEE Trans. Industrial Electronics and Control Instrumentation*, vol. IECI-25, no. 1, pp. 45–49, Feb. 1979.
12. Remy Simard and V. Rajagopalan, "Economical equidistant pulse firing scheme for thyristorized dc drives," *IEEE Trans. Industrial Electronics and Control Instrumentation*, vol. IECI-22, no. 3, pp. 425–429, Aug. 1975.
13. J. S. Wade Jr. and L. G. Aya, "Design for simultaneous pulse triggering of SCRs in three phase bridge configuration," *IEEE Trans. Industrial Electronics and Control Instrumentation*, vol. IECI-18, no. 3, pp. 104–106, Aug. 1971.
14. N. Peric and I. Petrovic, "Flying shear control system," *IEEE Trans. on Industry Applications*, vol. 26, no. 6, pp. 1049–1056, Nov./Dec. 1990.
15. Guy Lemieux, "Power system harmonic resonance—A documentary case," *IEEE Trans. on Industry Applications*, vol. 26, no. 3, pp. 483–488, May/June 1990.
16. Friedrich Frohr and Fritz Orttenger, *Introduction to Electronic Control Engineering*, Wiley Eastern Limited, Madras, 1988.
17. Leland A. Schlabach, "Analysis of discontinuous current in a 12-pulse thyristor dc motor drive," *IEEE Transactions on Industry Applications*, vol. 27, no. 6, pp. 1048–1054, Nov./Dec. 1991.
18. Hiromi Inaba, Seiya Shima, Akiteru Ueda, Takeki Ando, Toshiaki Kurosawa and Yoshio Sakai, "A new speed control system for dc motors using GTO converter and its application to elevators," *IEEE Transactions on Industry Applications*, vol. 21, no. 2, pp. 391–397, March/April 1985.
19. Geza Joos, "Variable Speed Drives," Short-term Course Lecture Notes, Center for Professional Development, NJ, Nov. 1987.

3.19 DISCUSSION QUESTIONS

1. The ratio between the field and armature time constants is of the order of 10 to 1000 in the separately-excited dc machines. Will it be of the same order in series field dc machines?
2. Single-phase converters are avoided in high-performance applications. Why?
3. Three-phase converters are better than single-phase converters when harmonics are considered. Will this advantage be of any use in motor drives?
4. Three-phase converters are faster in response than single-phase converters. Compare their speed of response.
5. The converters require a filter to tackle the harmonics. Which one of the following is preferable: A filter on the input side or one at the output side?
6. A freewheeling diode reduces the harmonics in the output current. This is true for triggering angles above a certain value. What is that limiting triggering angle?

7. Freewheeling can be accomplished without using a diode across the load by using the thyristors in the phase legs. Such a technique requires a modification of the triggering signals. Discuss the conceptual aspects of the modification.
8. Is a microprocessor necessary for implementing the modification in question 7?
9. Steady-state analysis of the dc motor drive assumes that there is no speed ripple. Is that justifiable?
10. Precise control of α is required for accurate control of a dc motor. How can this be ensured in practice?
11. A dc motor drive with an inner current loop alone is a torque amplifier. Is there any use for such a drive?
12. The speed-controlled dc drive uses actual speed for feedback control. In the absence of a tachogenerator, what would be done to close the speed loop?
13. Current feedback can be obtained by sensing the ac-side currents and rectifying them to find the armature current. What is the advantage of this method of current sensing over direct sensing of armature current?
14. Supply transformers are part of the dc motor-drive systems. Are they essential for all ranges of the motor drive?
15. Even though a dc motor-drive system is a sixth-order sampled-data control system, a number of approximations have been made to synthesize the controllers. Why not design the controllers without resorting to any approximations?
16. Can state feedback control be used in the design of current and speed controllers in the dc motor drive? Discuss the implementation of such a control scheme.
17. For speed-controller design, the fourth-order inner current loop has been approximated into a first-order transfer function. Discuss the merits and demerits of the approximation.
18. How critical is the current loop design in a two-/four-quadrant dc motor drive?
19. Discuss the design of the control and gating circuits in the four-quadrant dc motor drive.
20. The power switches have been assumed to be ideal, i.e., they have no forward voltage drops and losses. A dc motor drive is driven from a 12-V ac main. Can the switches be considered ideal now for the analysis of this dc motor drive?
21. Discuss the efficiency of the three-phase fully-controlled converter-fed dc motor drive.
22. "The phase-controlled converter is a harmonic generator." Justify this comment.
23. Load changes have a profound effect on the dynamic response of the dc motor drive. What are the techniques to counter the load sensitivity on the performance of the dc motor drive?

3.20 EXERCISE PROBLEMS

1. Derive the normalized steady-state performance equations of a series-excited dc motor drive.
2. A 100-hp open drip-proof dc motor rated at 500 V and 1750 rpm has the following parameters at rated field current.

$$\begin{aligned}
 I_a &= 153.71 \text{ A} & R_a &= 0.088 \, \Omega \\
 L_a &= 1.83 \text{ mH} & K_b &= 2.646 \text{ V/rad/sec}
 \end{aligned}$$

Draw its torque, induced emf, power, and field flux vs. speed in normalized units for rated armature current and for an intermittent operation at 1.2 p.u. armature current for the speed range of 0 to 2 p.u.

3. The dc motor given in Problem 2 is operated from a fully-controlled three-phase converter fed from a 460-V, 60-Hz, 3-phase ac main. Calculate the triggering angle when the machine is delivering rated torque at rated speed. The armature current is assumed to be continuous.
4. Assuming the current is continuous, draw α vs. speed, maintaining the load torque at rated value, for Problem 3, for a speed range of 0 to 1 p.u.
5. A separately-excited dc motor is controlled from a three-phase full-wave converter fed from a 460-V, 3-phase, 60-Hz ac supply. The dc motor details are as follows: 250 hp, 500 V, 1250 rpm, $R_a = 0.052 \Omega$, $L_a = 2$ mH.
 - (i) Find the rated current and K_b when the field is maintained at rated value.
 - (ii) Draw the torque-speed characteristics as a function of triggering angle, α .
6. The motor drive given in Example 3.1 is driving a load proportional to the square of the speed. Draw the torque-speed characteristics of the drive system and α vs. speed characteristics. The combined motor and load mechanical constants are as follows:
 $B_l = 0.06 \text{ N}\cdot\text{m}/\text{rad}/\text{sec}$, $J = 5 \text{ kg}\cdot\text{m}^2$
7. Determine the electrical and mechanical time constants of the dc motor in Problems 5 and 6, respectively.
8. The converter in one case is linearized and in the other case is not modified. The control voltage is $\pm 10 \text{ V}$ max in both cases. Design the current controller gains (PI) for the motor drive given in Problem 5. Consider $B_l = 0.06 \text{ N}\cdot\text{m}/\text{rad}/\text{sec}$ and $J = 5 \text{ kg}\cdot\text{m}^2$. Evaluate the time and frequency responses of the current loop for the two cases.
9. An innermost unity feedback voltage loop is introduced in Problem 8. The voltage controller is of proportional type, with gain K_v . Derive the current-loop transfer function, and explain the impact of the voltage loop on the performance of the current loop.
10. A two-quadrant dc motor drive is fed from a three-phase controlled converter operating on a 230-V, 60-Hz ac main. The motor details are as follows:

230 V, 40 hp, 1500 rpm.

$$\begin{array}{llll} R_a = 0.066 \Omega & L_a = 6.5 \text{ mH} & J = 25 \text{ kg}\cdot\text{m}^2 & B_l = 0.4 \text{ N}\cdot\text{m}/\text{rad}/\text{sec} \\ H_c = 0.05 \text{ V/A} & H_w = 0.576 \text{ V}/\text{rad}/\text{sec} & T_w = 0.002 \text{ sec} & K_b = 1.33 \text{ V}/\text{rad}/\text{sec} \end{array}$$

- The load is frictional. (i) Calculate the controller constants by using the method developed in the book. (ii) Evaluate the stability of the motor drive, using the complete model. (iii) Draw the Bode plots for the complete and simplified transfer functions between speed and its commanded value. (iv) Comment on the discrepancy in the results. (v) Select the ratings of the converter power switches. The maximum armature current is not to exceed 1.2 p.u.
11. The dc motor drive given in Example 3.5 has its inner current loop disabled and also its current controller removed. Compare its speed performance to the drive given in Example 3.5. In the present case, the speed controller constants will be different and hence will have to be recalculated.
 12. Prove that, when $T_c = T_1$, the damping coefficient and natural frequency of the current-loop transfer function remain equal to the case for $T_c = T_2$. Assume T_r is very, very small compared to T_1 and T_2 .
 13. Design gating and control circuitry for a half-controlled three-phase converter to drive a dc motor. Analyze the dc output voltage into harmonics and plot the dominant harmonic as a function of triggering angle.

14. A fully-controlled three-phase converter is driving a separately-excited dc motor in open loop. Plot its torque vs. speed characteristics as a function of triggering angle both in the continuous and in the discontinuous modes of operation. Use the motor parameters given in Problem 10.
15. The flying-shear control system described in the application section is to be designed with separately-excited dc motor drives. The power schematic of the drive system is shown in Figure 3.56(i); the control schematic is shown in Figure 3.56(ii). The system details are: Rated power = 1500 kW per motor, Rated voltage = 600 V per motor, Rated current = 2650 A per motor, Rated speed = 600 rpm. Armature resistance, $R_a = 0.003645 \Omega$ per motor, Armature inductance, $L_a = 0.0001$ H per motor, Emf constant, $K_b = 9.364$ V/rad/sec, Circulating current reactors, $L_c = 0.001$ H per motor, Total moment of inertia, $J = 1500 \text{ Kg-m}^2$, Gear ratio = 7, Radius of blade path = 0.75 m. Determine the controller gains and current-loop bandwidth if the input voltages supplied to the converter through an isolation transformer are 1000 V, 3-phase, 60 Hz. Assume 10 V equals 5 m/s and 10 V equals 2700 A for the control signals and that the metal input velocity is 4.2 m/s. Write a computer program to simulate the drive system and show the key variables as a function of time.
Refer to the paper 14 cited for relevant equations of the flying-shear control system.
16. Write a user-friendly computer program in a language of your choice to simulate the phase-controlled rectifier-fed separately-excited dc motor drive. The specifications of the drive system are as follows: The motor parameters are 150 hp, 220 V, 1500 rpm, base torque = 711.7 N·m, $R_a = 0.05 \Omega$, $L_a = 0.002$ H, $M I_t = K_b = 1.6$ V/rad/sec, $J = 0.16 \text{ kg-m}^2$, $B = 0.01$ N·m/rad/sec. The converter parameters are Supply input: 3-phase, 230 V, 60 Hz: Controller parameters: $V_{cm} = 10$ V, $H_w = 0.06364$ V/rad/sec, $T_w = 0.002$, $H_c = 0.01124$ V/A, $K_{pi} = 100$, $K_{ii} = 1.5$, $K_{ps} = 10$, $K_{is} = 1$, $T_{max} = 2 \times$ Base torque, $T_i = 0$ N·m. Plot the speed command, rotor speed, torque command, torque, current, current command, triggering angle, and applied voltage vs. time.

C H A P T E R 4

Chopper-Controlled DC Motor Drive

4.1 INTRODUCTION

In the case of single stage ac to dc power conversion phase-controlled converters described in Chapter 3 are used to drive the dc machines. Whenever the source is a constant-voltage dc, such as a battery or diode-bridge rectified ac supply, a different type of converter is required to convert the fixed voltage into a variable-voltage/variable-current source for the speed control of the dc motor drive. The variable dc voltage is controlled by chopping the input voltage by varying the on- and off-times of a converter, and the type of converter capable of such a function is known as a chopper.

The principle of operation of a four-quadrant chopper is explained in this chapter. The steady-state and dynamic analysis of the chopper-controlled dc motor drive and its performance characteristics are derived and evaluated. Some illustrative examples are included.

4.2 PRINCIPLE OF OPERATION OF THE CHOPPER

A schematic diagram of the chopper is shown in Figure 4.1. The control voltage to its gate is v_c . The chopper is on for a time t_{on} , and its off time is t_{off} . Its frequency of operation is

$$f_c = \frac{1}{(t_{on} + t_{off})} = \frac{1}{T} \quad (4.1)$$

and its duty cycle is defined as

$$d = \frac{t_{on}}{T} \quad (4.2)$$

The output voltage across the load during the on-time of the switch is equal to the difference between the source voltage V_s and the voltage drop across the power

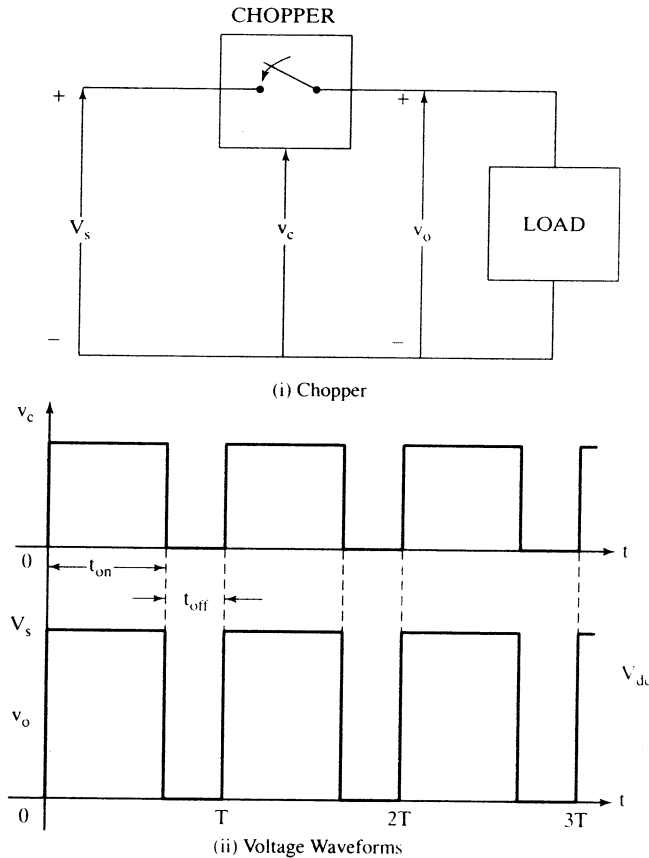


Figure 4.1 Chopper schematic and its waveforms

switch. Assuming that the switch is ideal, with zero voltage drop, the average output voltage V_{dc} is given as

$$V_{dc} = \frac{t_{on}}{T} V_s = dV_s \quad (4.3)$$

where V_s is the source voltage.

Varying the duty cycle changes the output voltage. Note that the output voltage follows the control voltage, as shown in Figure 4.1, signifying that the chopper is a voltage amplifier. The duty cycle d can be changed in two ways:

- (i) By keeping the switching/chopping frequency constant and varying the on-time, to get a changing duty cycle.
- (ii) Keeping the on-time constant and varying the chopping frequency, to obtain various values of the duty cycle.

A constant switching frequency has the advantages of predetermined switching losses of the chopper, enabling optimal design of the cooling for the power circuit, and predetermined harmonic contents, leading to an optimal input filter. Both of these advantages are lost by varying the switching frequency of the chopper; hence, this technique for chopper control is not prevalent in practice.

4.3 FOUR-QUADRANT CHOPPER CIRCUIT

A four-quadrant chopper with transistor switches is shown in Figure 4.2. Each transistor has a freewheeling diode across it and a snubber circuit to limit the rate of rise of the voltage. The snubber circuit is not shown in the figure.

The load consists of a resistance, an inductance, and an induced emf. The source is dc, and a capacitor is connected across it to maintain a constant voltage. The base drive circuits of the transistors are isolated, and they reproduce and amplify the control signals at the output. For the sake of simplicity, it is assumed that the switches are ideal and hence, the base drive signals can be used to draw the load voltage.

4.3.1 First-Quadrant Operation

First-quadrant operation corresponds to a positive output voltage and current. This is obtained by triggering T_1 and T_2 together, as is shown in Figure 4.3; then the load voltage is equal to the source voltage. To obtain zero load voltage, either T_1 or T_2 can be turned off. Assume that T_1 is turned off; then the current will decrease in the power switch and inductance. As the current tries to decrease in the inductance, it will have a voltage induced across it in proportion to the rate of fall of current with a polarity opposite to the load-induced emf, thus forward-biasing diode D_4 . D_4 provides the path for armature current continuity during this time. Because of this, the circuit configuration changes as shown in Figure 4.4. The load is short-circuited, reducing its voltage to zero. The current and voltage waveforms for continuous and

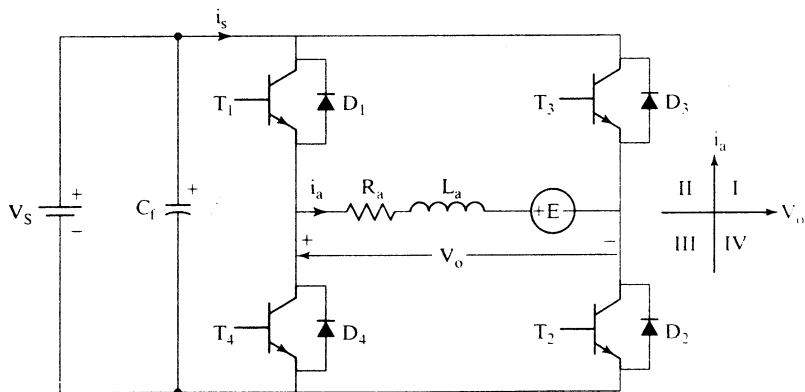


Figure 4.2 A four-quadrant chopper circuit

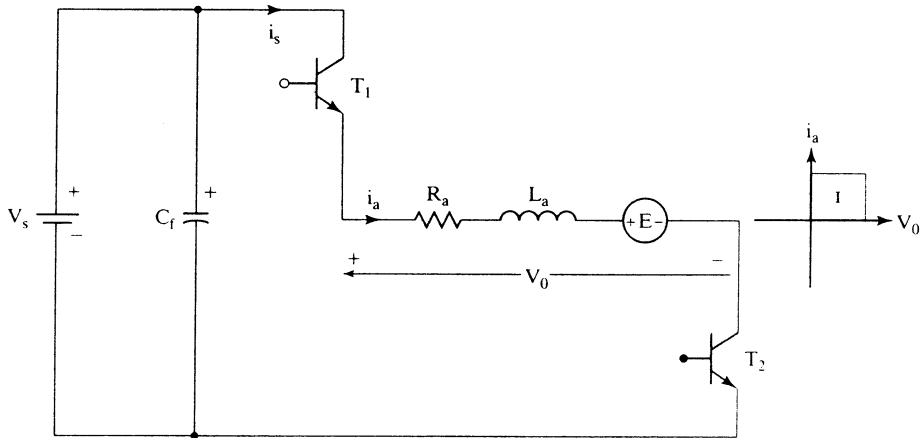


Figure 4.3 First-quadrant operation with positive voltage and current in the load

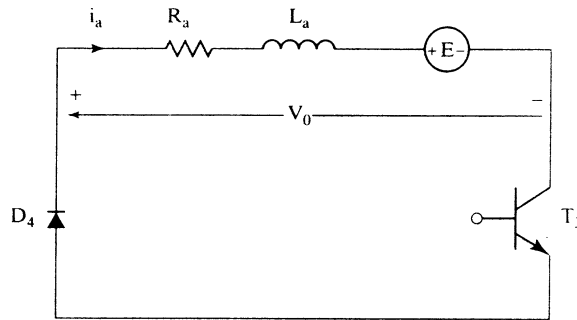
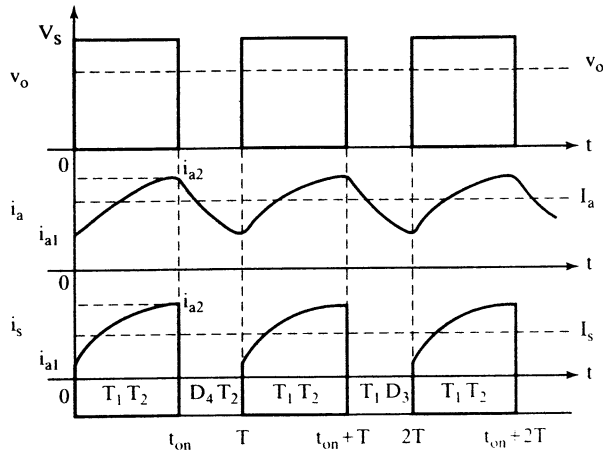


Figure 4.4 First-quadrant operation with zero voltage across the load

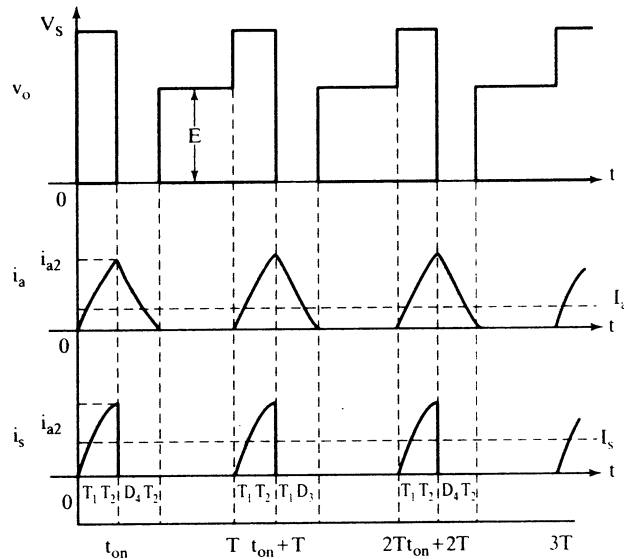
discontinuous current conduction are shown in Figure 4.5. Note that, in the discontinuous current-conduction mode, the induced emf of the load appears across the load when the current is zero. The load voltage, therefore, is a stepped waveform. The operation discussed here corresponds to motoring in the clockwise direction, or *forward motoring*. It can be observed that the average output voltage will vary from 0 to V_s ; the duty cycle can be varied only from 0 to 1.

The output voltage can also be varied by another switching strategy. Armature current is assumed continuous. Instead of providing zero voltage during turn-off time to the load, consider that T_1 and T_2 are simultaneously turned off, to enable conduction by diodes D_3 and D_4 . The voltage applied across the load then is equal to the negative source voltage, resulting in a reduction of the average output voltage. The disadvantages of this switching strategy are as follows:

- (i) Switching losses double, because two power devices are turned off instead of one only.
- (ii) The rate of change of voltage across the load is twice that of the other strategy. If the load is a dc machine, then it has the deleterious effect of causing



(i) Continuous Conduction



(ii) Discontinuous Conduction

Figure 4.5 Voltage and current waveforms in first-quadrant operation

higher dielectric losses in the insulation and therefore reduced life. Note that the dielectric is a capacitor with a resistor in series.

- (iii) The rate of change of load current is high, contributing to vibration of the armature in the case of the dc machine.
- (iv) Since a part of the energy is being circulated between the load and source in every switching cycle, the switching harmonic current is high, resulting in additional losses in the load and in the cables connecting the source and converter.

Therefore, this switching strategy is not considered any further in this chapter.

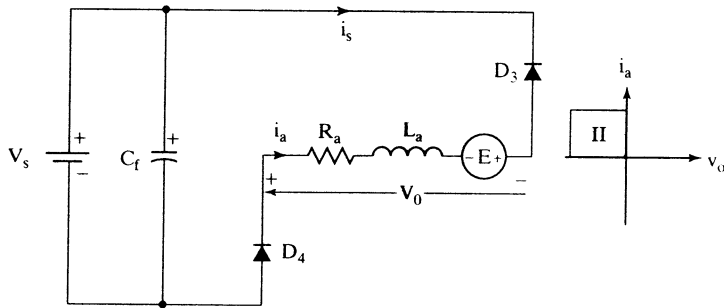


Figure 4.6 Second-quadrant operation, with negative load voltage and positive current

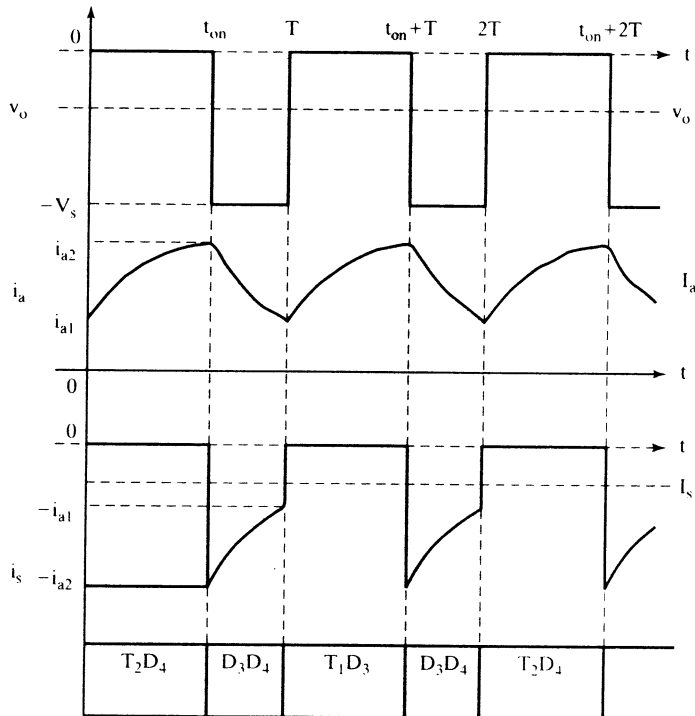


Figure 4.7 Second-quadrant operation of the chopper

4.3.2 Second-Quadrant Operation

Second-quadrant operation corresponds to a positive current with a negative voltage across the load terminals. Assume that the load's emf is negative. Consider that T_1 or T_2 is conducting at a given time. The conducting transistor is turned off. The current in the inductive load has to continue to flow until the energy in it is depleted to zero. Hence, the diodes D_3 and D_4 will take over, maintaining the load current in the same direction, but the load voltage is negative in the new circuit configuration, as is shown in Figure 4.6. The voltage and current waveforms are shown in Figure 4.7. When diodes D_3 and D_4 are conducting, the source receives power from the load. If the

source cannot absorb this power, provision has to be made to consume the power. In that case, the overcharge on the filter capacitor is periodically dumped into a resistor connected across the source by controlling the on-time of a transistor in series with a resistor. This form of recovering energy from the load is known as regenerative braking and is common in low-HP motor drives, where the saving in energy might not be considerable or cost-effective. When the current in the load is decreasing, T_2 is turned on. This allows the short-circuiting of the load through T_2 and D_4 , resulting in an increase in the load current. Turning off T_2 results in a pulse of current flowing into the source via D_3 and D_4 . This operation allows the priming up of the current and a building up of the energy in the inductor from the load's emf, thus enabling the transfer of energy from the load to the source. Note that it is possible to transfer energy from load to source even when E is lower in magnitude than V_c . This particular operational feature is sometimes referred to as *boost operation* in dc-to-dc power supplies. Priming up the load current can also be achieved alternatively, by using T_1 instead of T_2 .

4.3.3 Third-Quadrant Operation

Third-quadrant operation provides the load with negative current and voltage. A negative emf source, $-E$, is assumed in the load. Switching on T_3 and T_4 increases the current in the load, and turning off one of the transistors short-circuits the load, decreasing the load current. That way, the load current can be controlled within the externally set limits. The circuit configurations for the switching instants are shown in Figure 4.8. The voltage and current waveforms under continuous and discontinuous

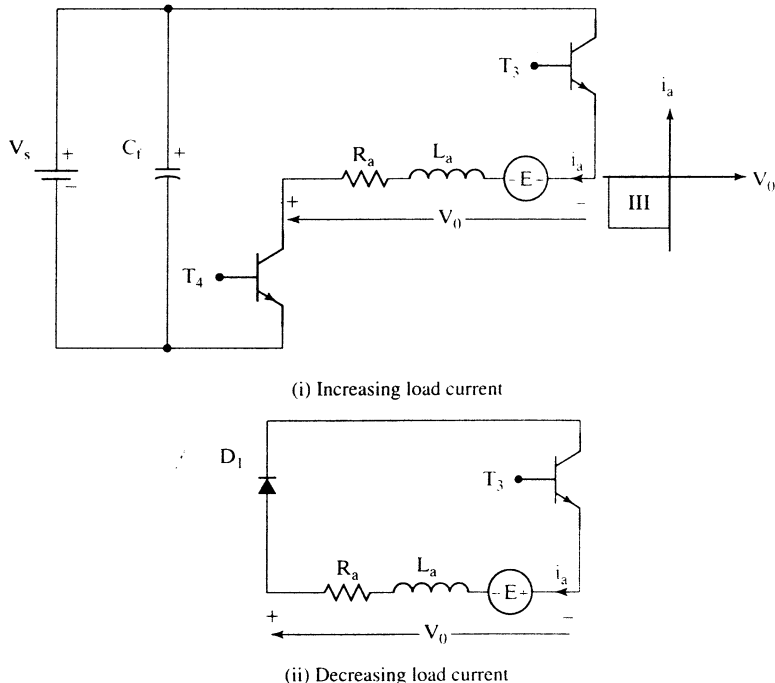


Figure 4.8 Modes of operation in the third quadrant

current-conduction modes are shown in Figure 4.9. Note the similarity between first- and third-quadrant operation.

4.3.4 Fourth-Quadrant Operation

Fourth-quadrant operation corresponds to a positive voltage and a negative current in the load. A positive load-emf source E is assumed. To send energy to the dc source from the load, note that the armature current has to be established to flow

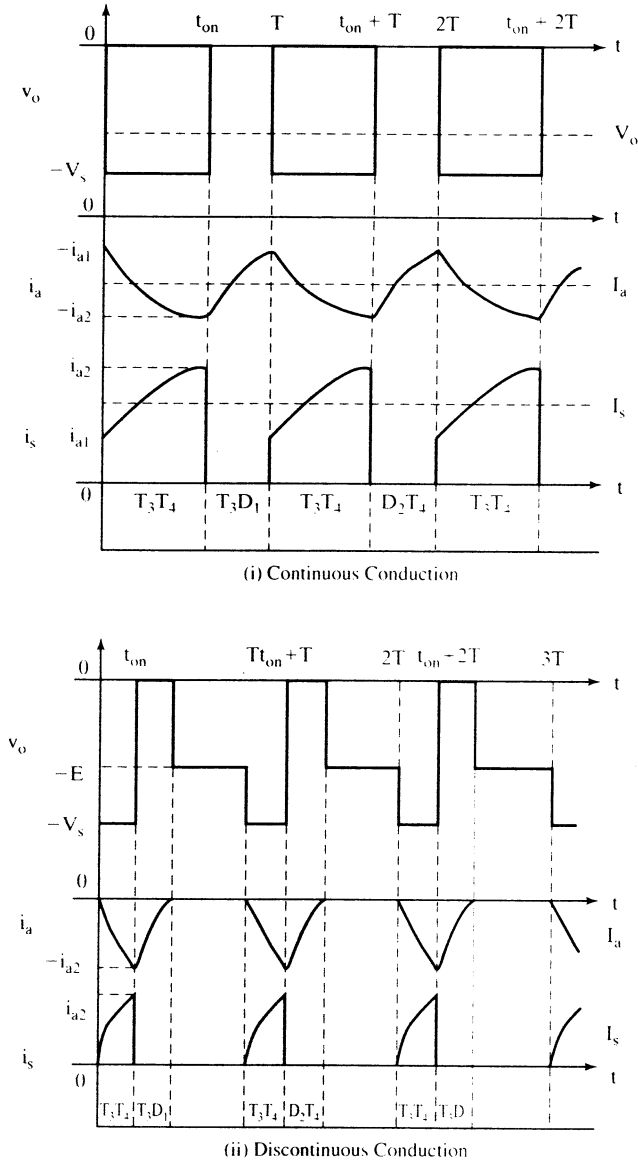


Figure 4.9 Third-quadrant operation

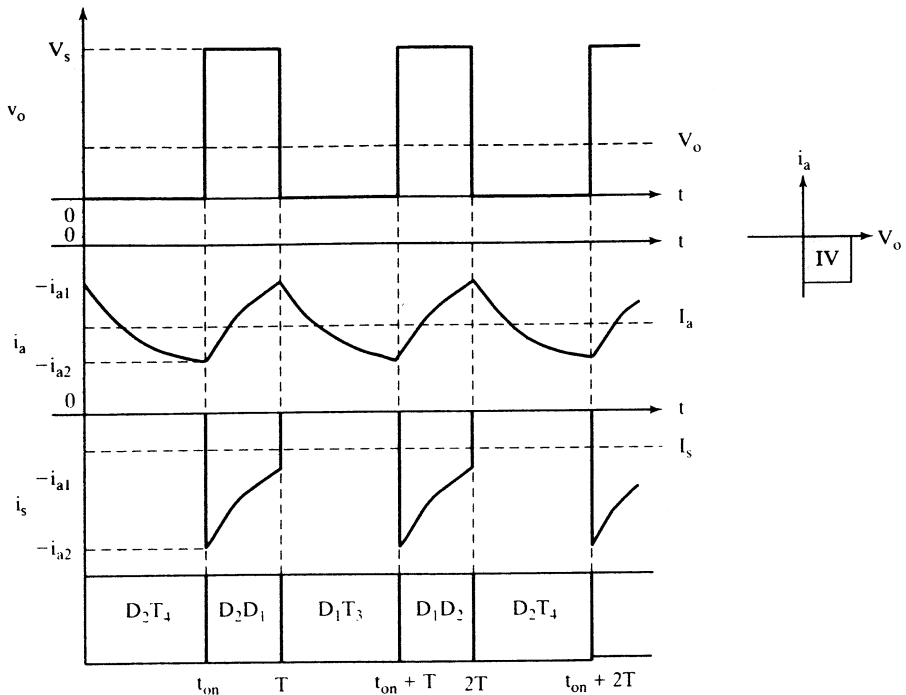


Figure 4.10 Fourth-quadrant operation of the chopper

from the right side to the left side as seen in Figure 4.2. By the convention adopted in this book, that direction of current is negative. Assume that the machine has been operating in quadrant I with a positive current in the armature. When a brake command is received, the torque and armature current command goes negative. The armature current can be driven negative from its positive value through zero. Opening T_1 and T_2 will enable D_3 and D_4 to allow current via the source, reducing the current magnitude rapidly to zero. To establish a negative current, T_4 is turned on. That will short-circuit the load, making the emf source build a current through T_4 and D_2 . When the current has reached a desired peak, T_4 is turned off. That forces D_1 to become forward-biased and to carry the load current to the dc input source via D_2 and the load. When the current falls below a lower limit, T_4 is again turned on, to build up the current for subsequent transfer to the source. The voltage and current waveforms are shown in Figure 4.10. The average voltage across the load is positive, and the average load current is negative, indicating that power is transferred from the load to the source. The source power is the product of average source current and average source voltage, and it is negative, as is shown in Figure 4.10.

4.4 CHOPPER FOR INVERSION

The chopper can be viewed as a single-phase inverter because of its ability to work in all the four quadrants, handling leading or lagging reactive loads in series with an emf source. The output fundamental frequency is determined by the rate at which

the forward-to-reverse operation is performed in the chopper. The chopper is the building block for a multiphase inverter.

4.5 CHOPPER WITH OTHER POWER DEVICES

Choppers are realized with MOSFETs, IGBTs, GTOs, or SCRs, depending upon the power level required. The MOSFET and transistor choppers are used at power levels up to 50 kW. Beyond that, IGBTs, GTOs, and SCRs are used for the power switches. Excepting SCR choppers, all choppers are self-commutating and hence have a minimum number of power switches and auxiliary components. In the case of SCR choppers, commutating circuits have to be incorporated for each main SCR. Such a circuit is described in Chapter 7.

4.6 MODEL OF THE CHOPPER

The chopper is modeled as a first-order lag with a gain of K_{ch} . The time delay corresponds to the statistical average conduction time, which can vary from zero to T . The transfer function is then

$$G_r(s) = \frac{K_r}{1 + \frac{sT}{2}} \quad (4.4)$$

where $K_r = V_s/V_{cm}$, V_s is the source voltage, and V_{cm} is the maximum control voltage.

Increasing the chopping frequency decreases the delay time, and hence the transfer function becomes a simple gain.

4.7 INPUT TO THE CHOPPER

The input to the chopper is either a battery or a rectified ac supply. The rectified ac is the prevalent form of input. The ac input is rectified through a diode bridge, and its output is filtered to keep the dc voltage a constant. The use of the diode bridge has the advantage of near-unity power factor, thus overcoming one of the serious disadvantages of the phase-controlled converter. It has a disadvantage: it cannot transfer power from the dc link into ac mains. In that case, the regenerative energy has to be dumped in the braking resistor, as shown in Figure 4.11, or a phase-controlled converter is connected antiparallel to the diode bridge to handle the regenerative energy, as shown in Figure 4.12. In the latter configuration, the phase-controlled converter can have a smaller rating than other power converters, because the rms value of the regenerative current will be small—the duration of regeneration is only a fraction of the motoring time for most loads.

The regenerating converter has to be operated at triggering angles greater than 90° , to reverse the output dc voltage across the converter to match the polarity of V_s . The phase-controlled converter is enabled only when V_s is greater than the

allowable magnitude ΔV over and above the nominal dc voltage obtained from the ac source. It is usual to set ΔV to be 15 or 20% of V . In such a case, there is a need for a step-up transformer in the path of the phase-controlled converter, to match the dc link voltage V_s . The phase converter is disabled when V_s is slightly greater than 1.35 V , where V is the line-to-line rms voltage, to prevent energy flow from source to dc link and from dc link to source via the phase-controlled converter.

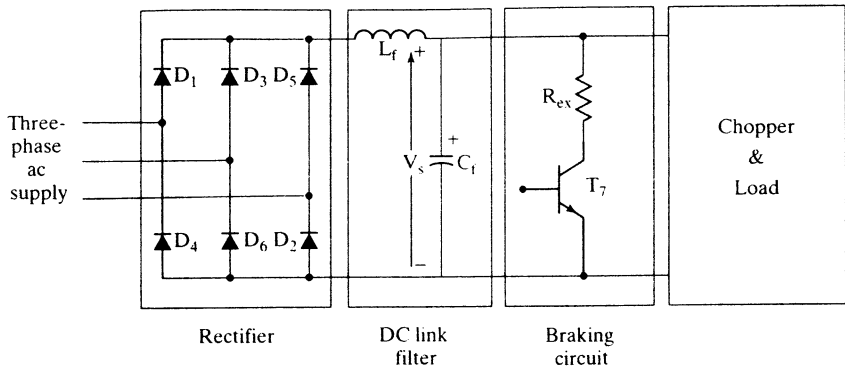


Figure 4.11 Front-end of the chopper circuit

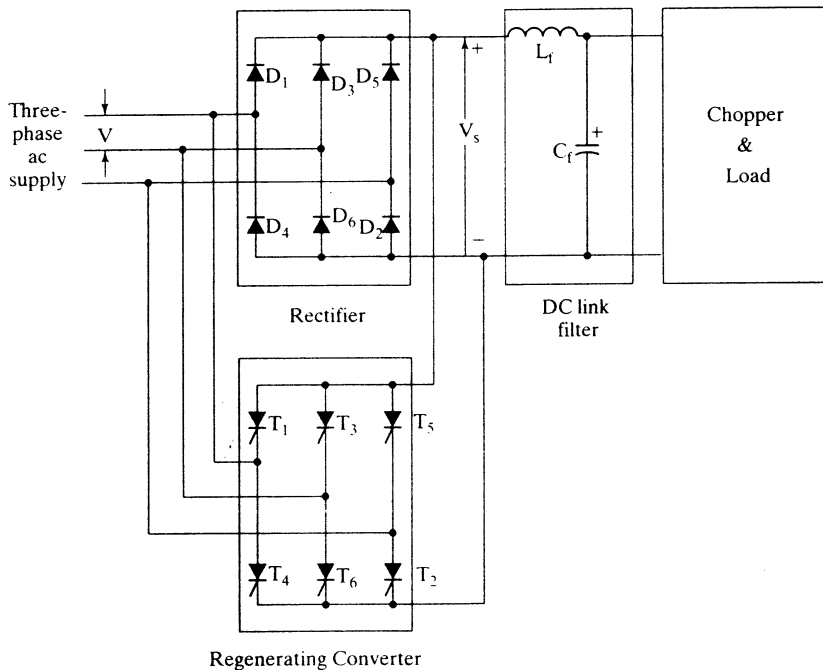


Figure 4.12 Chopper with regeneration capability

4.8 OTHER CHOPPER CIRCUITS

Not all applications demand four-quadrant operation. The power circuit is therefore simplified to accommodate only the necessary operation. Power devices are reduced for one- and two-quadrant drives, resulting in economy of the power converter. Apart from the chopper circuit shown in Figure 4.2 and its variations for reduced quadrants of operation, shown in Figure 4.13, a number of unique

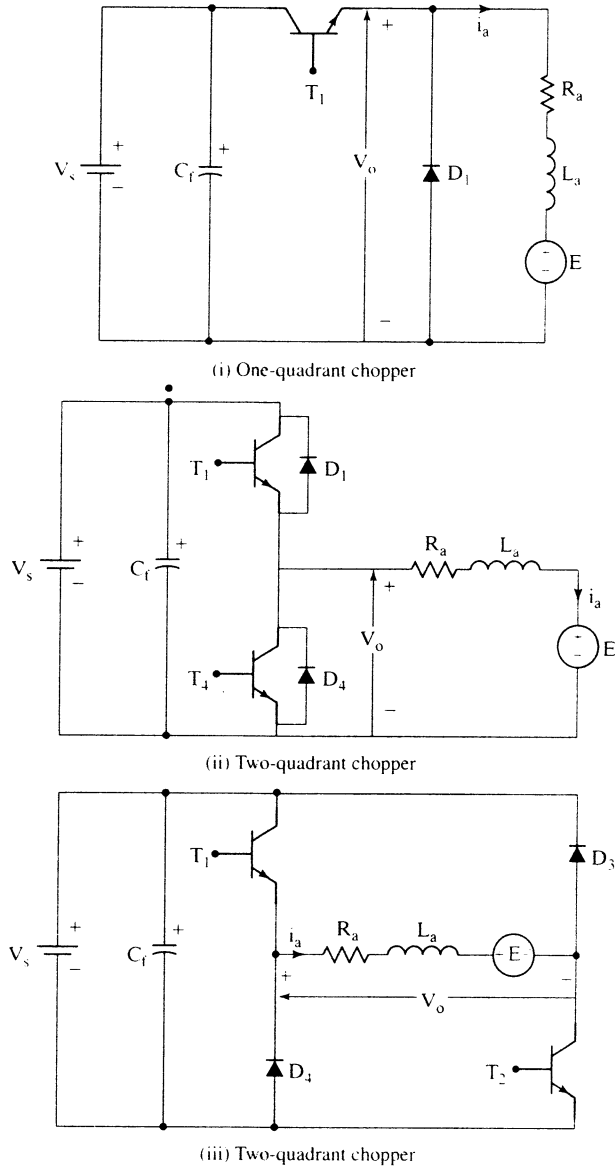


Figure 4.13 Variations of Figure 4.2 for one- and two-quadrant operation

chopper circuits are used. *Morgan* and *Jones* circuits are some of the commonly used choppers for dc motor speed control. They differ in their circuit topologies and commutation of the current from the circuit discussed in this chapter but they all have identical transfer characteristics. Considering this aspect, these circuits are not described and interested readers are referred to power-electronics textbooks.

4.9 STEADY-STATE ANALYSIS OF CHOPPER-CONTROLLED DC MOTOR DRIVE

The steady-state performance of the chopper-controlled dc motor drive is obtained either with average values, by neglecting harmonics, or including harmonics. The justification for using average values is that the average torque is the useful torque that is transmitted to the load. The torque components due to the current harmonics produce an average torque of zero over one cycle of switching. They do not contribute to useful power production. Further, they result in increased armature losses, because the harmonic currents increase the effective motor current. From an output point of view, neglecting harmonics and using only average values gives easier steady-state computation. The analysis by this method is known as *analysis by averaging*.

When losses, maximum steady-state current, and precise electromagnetic torque are required to fully analyze and design the drive system for an application, the true current waveforms in steady state need to be computed. Therefore, the harmonics cannot be excluded in the steady-state computation. A computationally efficient, analytical closed-form expression is obtained by the novel technique of boundary-matching conditions. This method is referred to as *instantaneous steady-state computation*.

It is assumed that the rotor speed is constant and the field is separately excited. For the following analysis, the field flux is maintained at rated value. For any other value of the field flux, the derivations need to be changed only with regard to the induced-emf term.

4.9.1 Analysis by Averaging

The average armature current is

$$I_{av} = \frac{V_{dc} - E}{R_a} \quad (4.5)$$

where

$$V_{dc} = dV_s \quad (4.6)$$

The electromagnetic torque is

$$T_{av} = K_b I_{av} \quad (4.7)$$

The torque, written in terms of duty cycle and speed from equations (4.5), (4.6) and (4.7), is

$$T_{av} = \frac{K_b(dV_s - K_b\omega_m)}{R_a} \text{ (N}\cdot\text{m)} \quad (4.8)$$

The electromagnetic torque is normalized by dividing it by the base torque, T_b , and by dividing both its numerator and denominator by the base voltage, V_b . The normalized torque is obtained by simplifying the expressions and substituting for V_b in terms of the base speed, ω , and emf constant:

$$T_{en} = \frac{T_{av}/V_b}{T_b/V_b} = \frac{K_b(dV_s - K_b\omega_m)/V_b}{K_bI_bR_a/V_b} = \frac{dV_n - \omega_{mn}}{R_{an}}, \text{ p.u.} \quad (4.9)$$

where

$$R_{an} = \frac{I_bR_a}{V_b}, \text{ p.u.} \quad (4.10)$$

$$V_n = \frac{V_s}{V_b}, \text{ p.u.} \quad (4.11)$$

$$\omega_{mn} = \frac{\omega_m}{\omega_b}, \text{ p.u.} \quad (4.12)$$

R_{an} , V_n , and ω_{mn} are normalized resistance, voltage, and speed, respectively. By assigning the product of normalized torque and p.u. resistance on the y axis, a set of normalized performance curves is drawn for various values of duty cycles, normalized voltage, and speed, as is shown in Figure 4.14. From the characteristics and the given duty cycle and voltage, the torque can be evaluated for a given speed, if the normalized resistance is known.

4.9.2 Instantaneous Steady-State Computation

The instantaneous steady-state armature current and electromagnetic torque including harmonics are evaluated in this section, both for continuous and discontinuous current conduction, by boundary-matching conditions. The relevant waveforms are shown in Figure 4.15. For each of the current-conduction modes, the performance is evaluated separately.

4.9.3 Continuous Current Conduction

The relevant electrical equations of the motor for on and off times are as follows:

$$V_s = E + R_a i_a + L_a \frac{di_a}{dt}, \quad 0 \leq t \leq dT \quad (4.13)$$

$$0 = E + R_a i_a + L_a \frac{di_a}{dt}, \quad dT \leq t \leq T \quad (4.14)$$

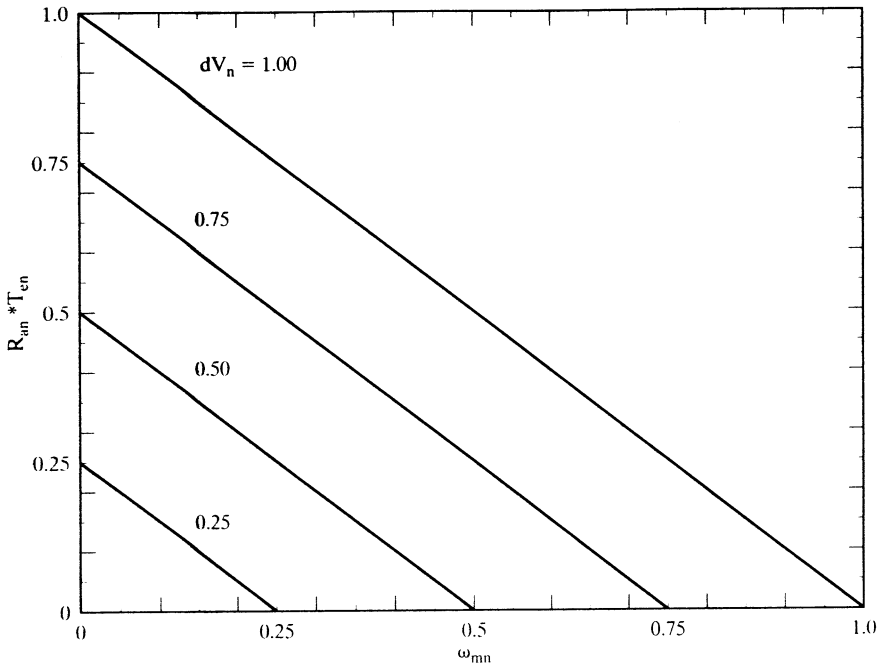


Figure 4.14 Normalized torque–speed characteristics as a function of duty cycle and source voltage

The solution of equation (4.13) is

$$i_a(t) = \frac{V_s - E}{R_a} (1 - e^{-\frac{t}{T_a}}) + I_{a0} e^{-\frac{t}{T_a}}, \quad 0 < t < dT \quad (4.15)$$

where

$$T_a = \text{Armature time constant} = \frac{L_a}{R_a} \quad (4.16)$$

Similarly, the solution of equation (4.14) is

$$i_a(t) = -\frac{E}{R_a} (1 - e^{-\frac{t}{T_a}}) + I_{a1} e^{-\frac{t}{T_a}}, \quad dT \leq t \leq dT \quad (4.17)$$

and

$$t^1 = t - dT \quad (4.18)$$

From Figure 4.15, it is seen that

$$i_a(t) = i_a(t + T) \quad (4.19)$$

By using this boundary condition, I_{a0} and I_{a1} are evaluated as

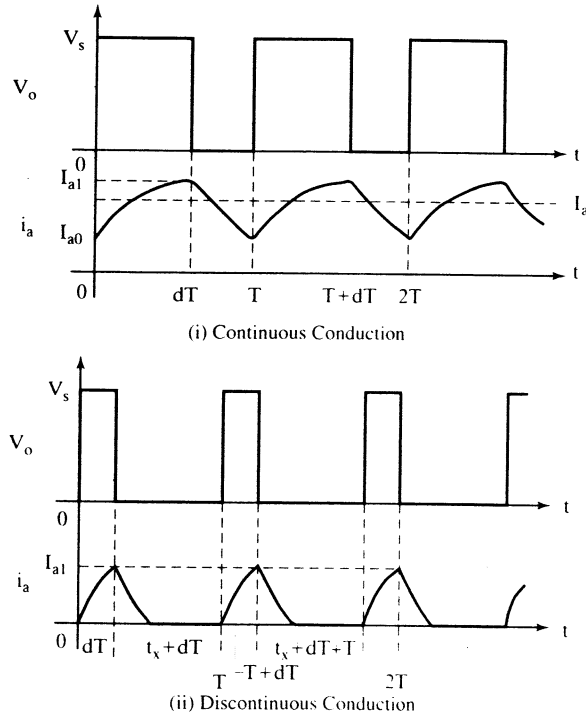


Figure 4.15 Applied voltage and armature current in a chopper-controlled dc motor drive

$$I_{a1} = \frac{V_s(1 - e^{-dT/T_a})}{R_a(1 - e^{-T/T_a})} - \frac{E}{R_a} \quad (4.20)$$

$$I_{a0} = \frac{V_s(e^{dT/T_a} - 1)}{R_a(e^{T/T_a} - 1)} - \frac{E}{R_a} \quad (4.21)$$

Having evaluated I_{a0} and I_{a1} , you can use equations (4.15) and (4.17) to evaluate the instantaneous armature current in steady state.

The limiting or minimum value of duty cycle for continuous current is evaluated by equating I_{a0} to zero. This value is termed the *critical duty cycle*, d_c , and is given by

$$d_c = \left(\frac{T_a}{T} \right) \log_e \left[1 + \frac{E}{V_s}(e^{T/T_a} - 1) \right] \quad (4.22)$$

Duty cycles lower than d_c will produce discontinuous current in the motor. Note that this critical value is dependent on the ratio between chopping time period and armature time constant and also on the ratio between induced emf and source voltage.

4.9.4 Discontinuous Current Conduction

The relevant equations for discontinuous current-conduction mode are obvious from Figure 4.15;

$$V_s = E + R_a i_a + L_a \frac{di_a}{dt}, \quad 0 < t < dT \quad (4.23)$$

$$0 = E + R_a i_a + L_a \frac{di_a}{dt}, \quad dT < t < (t_x + dT) \quad (4.24)$$

with

$$i_a(t_x + dT) = 0 \quad (4.25)$$

$$i_a(0) = 0 \quad (4.26)$$

Hence,

$$I_{a1} = \frac{V_s - E}{R_a} (1 - e^{-dT/T_a}) \quad (4.27)$$

$$i_a(t_x + dT) = -\frac{E}{R_a} (1 - e^{-\frac{t_x}{T_a}}) + I_{a1} e^{-\frac{t_x}{T_a}} \quad (4.28)$$

This equation is equal to zero, by the constraint given in equation (4.25), and, from that, t_x is evaluated as

$$t_x = T_a \log_e \left[1 + \frac{I_{a1} R_a}{E} \right] \quad (4.29)$$

The solution for the armature current in three time segments is

$$i_a(t) = \frac{V_s - E}{R_a} (1 - e^{-t/T_a}), \quad 0 < t < dT \quad (4.30)$$

$$i_a(t) = I_{a1} e^{-\frac{(t-dT)}{T_a}} - \frac{E}{R_a} \left(1 - e^{-\frac{(t-dT)}{T_a}} \right), \quad dT < t < t_x + dT \quad (4.31)$$

$$i_a(t) = 0, \quad (t_x + dT) < t < T \quad (4.32)$$

The steady-state performance is calculated by using equations (4.30) to (4.32).

Example 4.1

A dc motor is driven from a chopper with a source voltage of 24V dc and at a frequency of 1 kHz. Determine the variation in duty cycle required to have a speed variation of 0 to 1 p.u. delivering a constant 2 p.u. load. The motor details are as follows:

1 hp, 10 V, 2500 rpm, 78.5 % efficiency, $R_a = 0.01 \Omega$, $L_a = 0.002 \text{ H}$, $K_b = 0.03819 \text{ V/rad/sec}$

The chopper is one-quadrant, and the on-state drop voltage across the device is assumed to be 1 V regardless of the current variation.

Solution (i) Calculation of rated and normalized values

$$V_b = 10 \text{ V}$$

$$V_n = \frac{V_s}{V_b} = \frac{24 - 1}{10} = 2.3 \text{ p.u.}$$

$$\omega_{mr} = \frac{2500 \times 2\pi}{60} = 261.79 \text{ rad/sec}$$

$$I_{ar} = \frac{\text{Output}}{\text{Voltage} \times \text{Efficiency}} = \frac{1 \times 746}{10 \times 0.785} = 95 \text{ A} = I_b$$

$$R_{an} = \frac{I_b R_a}{V_b} = \frac{95 \times 0.001}{10} = 0.095 \text{ p.u.}$$

$$T_{en} = 2 \text{ p.u.}$$

(ii) Calculation of duty cycle

The minimum and maximum duty cycles occur at 0 and 1 p.u. speed, respectively, and at 2 p.u. load. From equation (4.9),

$$d = \frac{T_{en} R_{an} + \omega_{mn}}{V_n}$$

$$d_{\min} = \frac{2 \times 0.095 + 0}{2.3} = 0.0826$$

$$d_{\max} = \frac{2 \times 0.095 + 1}{2.3} = 0.517$$

The range of duty cycle variation required, then, is

$$0.0826 \leq d \leq 0.517$$

Example 4.2

The critical duty cycle can be changed by varying either the electrical time constant or the chopping frequency in the chopper. Draw a set of curves showing the effect of these variations on the critical duty cycle for various values of E/V_s .

Solution

$$d_c = \left(\frac{T_a}{T} \right) \log_e \left[1 + \frac{E}{V_s} \left(e^{\frac{T}{T_a}} - 1 \right) \right]$$

In terms of chopping frequency,

$$d_c = f_c T_a \log_e \left[1 + \frac{E}{V_s} \left(e^{\frac{1}{f_c T_a}} - 1 \right) \right]$$

Assigning various values of E/V_s and varying $f_c T_a$ would yield a set of critical duty cycles. The graph between d_c and $f_c T_a$ for varying values of E/V_s is shown in Figure 4.16. The maximum value of $f_c T_a$ is chosen to be 10.

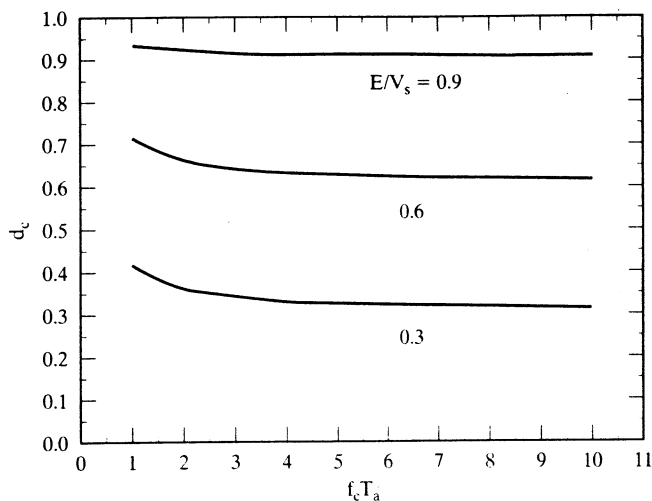


Figure 4.16 Critical duty cycles vs. the product of chopping frequency and electrical time constant of the dc motor, as a function of induced emf and source voltage

Example 4.3

A 200-hp, 230-V, 500-rpm separately-excited dc motor is controlled by a chopper. The chopper is connected to a bridge-diode rectifier supplied from a 230-V, 3- ϕ , 60-Hz ac main. The motor chopper details are as follows:

$$R_a = 0.04 \, \Omega, L_a = 0.0015 \, \text{H}, K_b = 4.172 \, \text{V/rad/sec}, f_c = 2 \, \text{kHz}.$$

The motor is running at 300 rpm with 55% duty cycle in the chopper. Determine the average current from steady-state current waveform and the electromagnetic torque produced in the motor. Compare these results with those obtained by averaging.

Solution The critical duty cycle is evaluated to determine the current continuity at the given duty cycle of 0.55.

$$d_c = \left(\frac{T_a}{T} \right) \log_e \left[1 + \frac{E}{V_s} (e^{\frac{T}{T_a}} - 1) \right]$$

$$T_a = \frac{0.0015}{0.04} = 0.0375 \, \text{s}$$

$$T = \frac{1}{f_c} = \frac{1}{2 \times 10^3} = 0.5 \, \text{ms}$$

$$\frac{T_a}{T} = 75$$

$$V_s = 1.35 \, \text{V} \cos \alpha = 1.35 \times 230 \times \cos 0^\circ = 310.5 \, \text{V}$$

$$E = K_b \omega_m = 4.172 \times \frac{2\pi \times 300}{60} = 131.1 \, \text{V}$$

$$d_c = 75 \log_e \left[1 + \frac{131.1}{310.5} (e^{\frac{1}{75}} - 1) \right] = 0.423$$

The given value of d is greater than the critical duty cycle; hence, the armature current is continuous.

$$I_{a0} = \frac{V_s(e^{dT/T_s} - 1)}{R_a(e^{T/T_s} - 1)} - \frac{E}{R_a} = \frac{310.5(e^{0.55/75} - 1)}{0.04(e^{1/75} - 1)} - \frac{131.1}{0.04} = 979 \text{ A}$$

$$I_{a1} = \frac{V_s(1 - e^{-dT/T_s})}{R_a(1 - e^{-T/T_s})} - \frac{E}{R_a} = \frac{310.5(1 - e^{-0.55/75})}{0.04(1 - e^{-1/75})} - \frac{131.1}{0.04} = 1004.7 \text{ A}$$

The average current is

$$I_{av} = \frac{1}{T} \left[\int_0^{dT} \left(\frac{V_s - E}{R_a} (1 - e^{-t/T_s}) + I_{a0} e^{-t/T_s} \right) dt + \int_0^{(1-d)T} \left(-\frac{E}{R_a} (1 - e^{-t/T_s}) + I_{a1} e^{-t/T_s} \right) dt \right]$$

$$= \frac{1}{T} \left[\frac{V_s - E}{R_a} \{dT + T_s(e^{-dT/T_s} - 1)\} + I_{a0} T_s(1 - e^{-dT/T_s}) - \frac{E}{R_a} \{(1-d)T - T_s + T_s e^{-(1-d)T/T_s}\} + I_{a1} T_s(1 - e^{-(1-d)T/T_s}) \right] = 991.8 \text{ A}$$

$$T_{av} = K_b I_{av} = 4.172 \times 991.8 = 4137.7 \text{ N}\cdot\text{m}$$

Steady state by averaging

$$I_{av} = \frac{(dV_s - K_b \omega_m)}{R_a} = \frac{0.55 \times 310.5 - 4.172 \times 31.42}{0.04} = 991.88 \text{ A}$$

$$T_{av} = K_b I_{av} = 4138.1 \text{ N}\cdot\text{m}$$

There is hardly any significant difference in the results by these two methods.

4.10 RATING OF THE DEVICES

The chopper shown in Figure 4.2 is considered, for illustration. The armature current is assumed to be continuous, with no ripples. If $I_{\max}$ is the maximum allowable current in the dc machine, the rms value of the power switch is dependent on this value and its duty cycle. The duty cycle of one device is slightly more than the duty cycle of the chopper, because one of the power switches continues to carry current during freewheeling while the other is turned off. In order to equal the current loading of the switches, the power switch carrying the freewheeling current is turned off in the next cycle and the previously inactive switch is allowed to carry the freewheeling current alternately. Accordingly, the current waveform of one power switch and the diode is as shown in Figure 4.17. Note that the diodes are, like their respective power switches, alternately carrying the freewheeling current, and only motoring action is considered for the calculation.

The rms value of the power switch current and average diode current are given by,

$$I_t = \sqrt{\frac{I_{\max}^2}{2T}(T + dT)} = \sqrt{\frac{1+d}{2}} \cdot I_{\max} \quad (4.33)$$

$$I_d = \left(\frac{1-d}{2} \right) \cdot I_{\max} \quad (4.34)$$

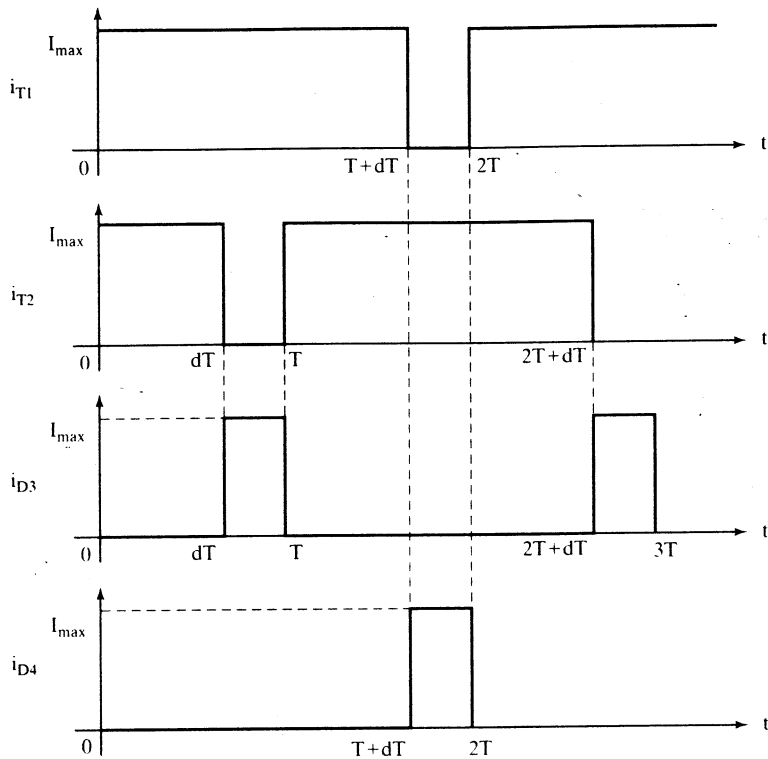


Figure 4.17 Transistor and diode currents in the chopper for motoring operation in the continuous-conduction mode

where the subscripts d and T refer to the diode and power switch, respectively. The average diode and rms power switch currents vs. duty cycle for continuous conduction are shown in Figure 4.18, for design use.

The minimum voltage rating for both the devices is,

$$V_T = V_d = V_s \quad (4.35)$$

Whenever regeneration occurs in the motor drive, the current in the freewheeling diodes would change, and, depending on the frequency and duration of regeneration, the diode currents have to be recalculated. To optimize the chopper rating in comparison to the motor and load demands, the operating conditions have to be known beforehand. The extreme operating conditions would then prevail on the design and hence on the final rating of the chopper.

4.11 PULSATING TORQUES

The armature current has ac components. These ac components or harmonics produce corresponding pulsating torques. The average of the harmonic torques is zero,

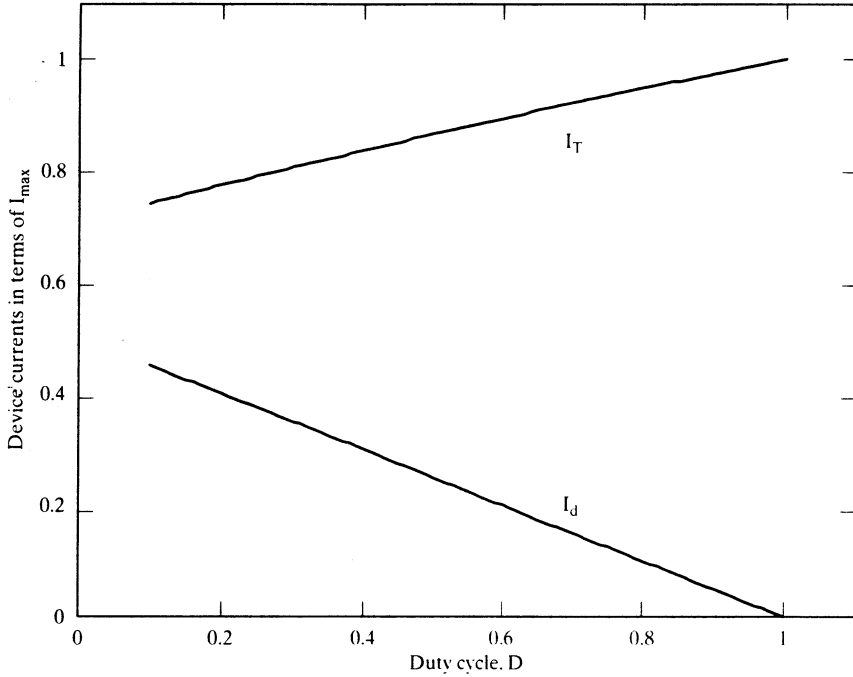


Figure 4.18 Device currents vs. duty cycle for continuous-conduction mode

and they do not contribute to useful torque and power. Some high-performance applications, such as machine tools and robots, require the pulsating torque to be a minimum so as not to degrade the process and the products. In that case, an estimation of the pulsating torques is in order.

The pulsating currents are evaluated from the harmonic voltages and the harmonic armature impedances of the dc machine. The applied voltage shown in Figure 4.10 is resolved into Fourier components as

$$v_a(t) = V_a + \sum_{n=1}^{\infty} A_n \sin(n\omega_c t + \theta_n) \quad (4.36)$$

where

$$V_a = \frac{dT}{T} \cdot V_s = dV_s, \quad V \quad (4.37)$$

$$\omega_c = 2\pi f_c = \frac{2\pi}{T}, \quad \text{rad/sec} \quad (4.38)$$

$$A_n = \frac{2V_s}{n\pi} \sin \frac{n\omega_c dT}{2}, \quad V \quad (4.39)$$

$$\theta_n = \frac{\pi}{2} - \frac{n\omega_c dT}{2}, \quad \text{rad} \quad (4.40)$$

and n is the order of the harmonic. The armature current is expressed as

$$i_a(t) = I_{av} + \sum_{n=1}^{\infty} \frac{A_n}{|Z_n|} \sin(n\omega_c t + \theta_n - \phi_n) \quad (4.41)$$

where

$$I_{av} = \frac{V_a - E}{R_a} \quad (4.42)$$

$$Z_n = R_a + jn\omega_c L_a \quad (4.43)$$

$$\phi_n = \cos^{-1} \left\{ \frac{R_a}{\sqrt{R_a^2 + n^2 \omega_c^2 L_a^2}} \right\} \quad (4.44)$$

and Z_{an} is the harmonic impedance of the armature. The input power is

$$P_i = v_a(t) i_a(t) = \left\{ V_a I_{av} + I_{av} \sum_{n=1}^{\infty} A_n \sin(n\omega_c t + \theta_n) + V_a \sum_{n=1}^{\infty} \frac{A_n}{|Z_n|} \sin(n\omega_c t + \theta_n - \phi_n) + \sum_{n=1}^{\infty} \left(\frac{A_n^2}{|Z_n|} \sin(n\omega_c t + \theta_n) \sin(n\omega_c t + \theta_n - \phi_n) \right) \right\} \quad (4.45)$$

The right-hand side of equation (4.45) can be simplified further, because the following pulsating terms reduce to an average of zero.

$$I_{av} \sum_{n=1}^{\infty} A_n \sin(n\omega_c t + \theta_n) = 0 \quad (4.46)$$

$$V_a \sum_{n=1}^{\infty} \frac{A_n}{|Z_n|} \sin(n\omega_c t + \theta_n - \phi_n) = 0 \quad (4.47)$$

The other terms tend to zero, as is shown below.

$$\begin{aligned} & \sum_{n=1}^{\infty} \left(\frac{A_n^2}{|Z_n|} \sin(n\omega_c t + \theta_n) \sin(n\omega_c t + \theta_n - \phi_n) \right) \\ &= \sum_{n=1}^{\infty} \left(\frac{A_n^2}{|Z_n|} \cdot \frac{1}{2} \{ \cos \phi_n - \cos(2n\omega_c t + 2\theta_n - \phi_n) \} \right) \end{aligned} \quad (4.48)$$

The average of the double-frequency term is zero, and the power factor is nearly zero because

$$n^2 \omega_c^2 L_a^2 \gg R_a \quad (4.49)$$

Hence, the expression (4.48) is almost equal to zero. The average input power becomes a constant quantity and is expressed as

$$P_{av} = V_a I_{av} \quad (4.50)$$

Power calculation can use the average values, and there is no need to resort to harmonic analysis, as is shown by the above derivation. The fundamental-harmonic peak pulsating torque then is

$$T_{cl} = K_b i_{al} \quad (4.51)$$

and the fundamental-harmonic armature current is given by

$$i_{a1} = \frac{2V_s}{\pi\sqrt{R_a^2 + \omega_c^2 L_a^2}} \sin\left(\frac{\omega_c d T}{2}\right) \quad (4.52)$$

The fundamental armature current can be alternately expressed in terms of duty cycle as

$$i_{a1} = \frac{2V_s}{\pi\sqrt{R_a^2 + \omega_c^2 L_a^2}} \sin(\pi d) \quad (4.53)$$

When the duty cycle is 50%, the fundamental armature current, and hence the pulsating torque, is maximum. For a duty cycle of 100%, there are no pulsating-torque components. The instantaneous electromagnetic torque is the sum of the dc and harmonic torques, written as

$$T_c(t) = T_{av} + T_{ch} \quad (4.54)$$

where h is the harmonic order. The harmonic torques, T_{ch} , do not contribute to the load: their averages are zero.

The fundamental pulsating torque is expressed as a fraction of average torque, to examine the impact of duty cycle on the pulsating torque. It is facilitated by the following development.

$$\begin{aligned} \frac{T_{el}}{T_{av}} &= \frac{K_b i_{a1}}{K_b I_{av}} = \frac{2V_s \sin(\pi d) / (\pi\sqrt{R_a^2 + \omega_c^2 L_a^2})}{(dV_s - E)/R_a} = \frac{2R_a}{\pi\sqrt{R_a^2 + \omega_c^2 L_a^2}} \cdot \frac{V_s}{(dV_s - E)} \sin(\pi d) \quad (4.55) \\ &= \left(\frac{2}{\pi} \cos \phi_1\right) \frac{\sin(\pi d)}{\left(d - \frac{E}{V_s}\right)} = k_1 \cdot \frac{\sin(\pi d)}{\left(d - \frac{E}{V_s}\right)} \end{aligned}$$

where ϕ_1 is the fundamental power-factor angle, which can be extracted from

$$\cos \phi_1 = \frac{R_a}{\sqrt{R_a^2 + \omega_c^2 L_a^2}} \quad (4.56)$$

and

$$k_1 = \frac{2}{\pi} \cos \phi_1 \quad (4.57)$$

A set of normalized curves is shown in Figure 4.19 to indicate the influence of duty cycle and of the ratio between the induced emf and the source voltage on the magnitude of the pulsating torque.

By noting that the fundamental is the predominant component among the ac components, the rms value of the armature current is approximated as

$$I_{rms} = \sqrt{I_{av}^2 + I_1^2} \quad (4.58)$$

where I_1 is the fundamental rms current, given by

$$I_1 = \frac{i_{a1}}{\sqrt{2}} = \frac{\sqrt{2}V_s}{\pi\sqrt{R_a^2 + \omega_c^2 L_a^2}} \sin(\pi d) \quad (4.59)$$

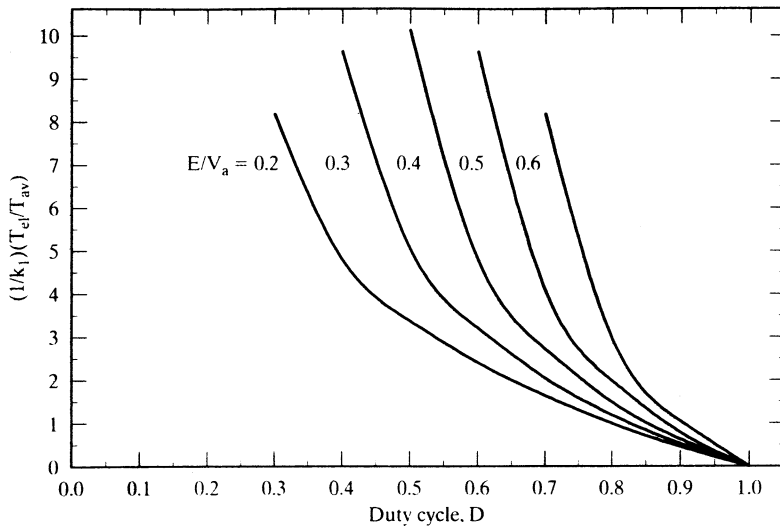


Figure 4.19 Normalized fundamental torque pulsation vs. duty cycle as a function of the ratio between induced emf and source voltage

The armature resistive losses are,

$$P_c = I_{\text{rms}}^2 R_a \quad (4.60)$$

This indicates that the thermal capability of the motor is degraded by the additional losses produced by the harmonic currents.

Minimization of the dominant-harmonic torque is of importance in many applications, mainly in positioning of machine tool drives. The key to mitigation of the harmonic torque is revealed by the expression for harmonic current given by equation (4.53). Given a fixed voltage source, there are only two variables that could be utilized to reduce harmonic current: the chopping frequency, and the machine inductance. Note that d cannot be used; it is a variable dependent on the speed and load. The chopping frequency is limited by the selection of power device, by its switching losses, and by other factors, such as electromagnetic compatibility. The advantage of varying the carrier frequency is that it is machine-independent, and hence the solution is contained within the chopper. This may not be feasible, as in the case of large (>100 hp) motor drives, but it is possible to increase the armature inductance of the machine during the design of the motor or to include an external inductor to increase the effective inductance in the armature path. The latter solution is the only practical approach in retrofit applications.

Example 4.4

A separately-excited dc motor is controlled by a chopper whose input dc voltage is 180 V. This motor is considered for low-speed applications requiring less than 2% pulsating torque at 300 rpm. (i) Evaluate its suitability for that application. (ii) If it is found unsuit-

able, what is the chopping frequency that will bring the pulsating torque to the specification? (iii) Alternatively, a series inductor in the armature can be introduced to meet the specification. Determine the value of that inductor. The motor and chopper data are as follows:

$$3 \text{ hp, } 120 \text{ V, } 1500 \text{ rpm, } R_a = 0.8 \Omega, L_a = 0.003 \text{ H, } K_b = 0.764 \text{ V/rad/sec, } f_c = 500 \text{ Hz}$$

Solution Rated torque,

$$T_{er} = \frac{3 \times 745.6}{2\pi \times 1500/60} = 14.25 \text{ N}\cdot\text{m}$$

$$\text{Maximum pulsating torque permitted} = 0.02 \times T_{er} = 0.02 \times 14.25 = 0.285 \text{ N}\cdot\text{m}$$

To find the harmonic currents, it is necessary to know the duty cycle. That is approximately determined by the averaging-analysis technique, by assuming that the motor delivers rated torque at 300 rpm.

$$I_b = \frac{T_{er}}{K_b} = \frac{14.25}{0.764} = 18.65 \text{ A}$$

$$V_a = E + I_{av}R_a = K_b\omega_m + I_{av}R_a = 0.764 \times \frac{2\pi \times 300}{60} + 18.65 \times 0.8 = 38.91 \text{ V}$$

$$d = \frac{V_a}{V_s} = \frac{38.91}{180} = 0.216$$

(i) The fundamental pulsating torque is assumed to be predominant for this analysis.

$$i_{a1} = \frac{2V_s}{\pi\sqrt{R_a^2 + \omega_c^2 L_a^2}} \sin(\pi d) = \frac{2 \times 180}{\pi\sqrt{0.8^2 + (2\pi \times 500 \times 0.003)^2}} \sin(0.216\pi) = 7.6 \text{ A.}$$

$$T_{c1} = K_b i_{a1} = 0.764 \times 7.6 = 5.8 \text{ N}\cdot\text{m}$$

This pulsating torque exceeds the specification, and, hence, in the present condition, the drive is unsuitable for use.

(ii) The fundamental current to produce 2% pulsating torque is

$$i_{a1(\text{spec})} = \frac{T_{c1(\text{spec})}}{K_b} = \frac{0.285}{0.764} = 0.373 \text{ A}$$

$$i_{a1(\text{spec})} = \frac{2V_s}{\pi\sqrt{R_a^2 + \omega_c^2 L_a^2}} \sin(\pi d)$$

from which the angular switching frequency to meet the fundamental current specification is obtained as

$$\omega_{c1} = \sqrt{\frac{4V_s^2 \sin^2(\pi d)}{\pi^2 L_a^2 i_{a1(\text{spec})}^2} - \frac{R_a^2}{L_a^2}} = \sqrt{(4 \times 180^2 \sin^2(0.216\pi)/(\pi^2(0.003)^2(0.373)^2) - \left(\frac{0.8}{0.003}\right)^2}$$

$$= 64.278 \text{ rad/s}$$

$$f_{c1} = \frac{\omega_{c1}}{2\pi} = 10.23 \text{ kHz}$$

Note that f_{cl} is the chopping frequency in Hz, which decreases the pulsating torque to the specification.

(iii) Let L_{ex} be the inductor introduced in the armature circuit. Then its value is

$$L_{ex} = \sqrt{\frac{4V_s^2 \sin^2(\pi d)}{\pi^2 \omega_c^2 I_{al(spec)}^2} - \frac{R_a^2}{\omega_c^2}} - L_a$$

$$= \sqrt{(4 \times 180^2 \sin^2(0.216\pi)) / (\pi^2 (2\pi \times 500)^2 (0.373)^2) - \left(\frac{0.8}{2\pi \times 500}\right)^2} - 0.003 = 71.5 \text{ mH}$$

Example 4.5

Calculate (i) the maximum harmonic resistive loss and (ii) the derating of the motor drive given in Example 4.4. The motor is operated with a base current of 18.65 A, which is inclusive of the fundamental-harmonic current. Consider only the dominant-harmonic component, to simplify the calculation.

Solution

$$I_{rms}^2 = I_b^2 = I_{av}^2 + I_1^2$$

By dividing by the square of the base current, the equation is expressed in terms of the normalized currents as

$$I_{avn}^2 + I_{1n}^2 = 1 \text{ p.u.}$$

where

$$I_{1n} = \frac{\sqrt{2}}{\pi} \frac{V_s}{\sqrt{R_a^2 + \omega_c^2 L_a^2}} \sin(\pi d) \frac{1}{I_b} = \frac{\sqrt{2}}{\pi} \frac{V_{sn}}{Z_{an}} \sin(\pi d)$$

where

$$V_{sn} = \frac{V_s}{V_b}; Z_{an} = \frac{Z_a}{Z_b} = \frac{\sqrt{R_a^2 + \omega_c^2 L_a^2}}{Z_b}$$

$$Z_b = \frac{V_b}{I_b} = \frac{120}{18.65} = 6.43 \Omega$$

$$V_{sn} = \frac{180}{120} = 1.5 \text{ p.u.}$$

$$Z_a = \sqrt{0.8^2 + (2\pi * 500 * 0.003)^2} = 9.42 \Omega$$

$$Z_{an} = \frac{9.42}{6.43} = 1.464 \text{ p.u.}$$

For a duty cycle of 0.5, the dominant-harmonic current is maximum and is given as

$$I_{1n} = \frac{\sqrt{2}}{\pi} \frac{1.5}{1.464} = 0.46 \text{ p.u.}$$

(i) The dominant-harmonic armature resistive loss is

$$P_{1n} = I_{1n}^2 R_{an} = 0.46^2 \frac{0.8}{6.43} = 0.02628 \text{ p.u.}$$

- (ii) For equality of losses in the machine with pure and chopped-current operation, the average current in the machine with chopped-current operation is derived as

$$I_{avn} = \sqrt{1 - I_{in}^2} = \sqrt{1 - .46^2} = 0.887 \text{ p.u.}$$

which translates into an average electromagnetic torque of 0.887 p.u., resulting in 11.3% derating of the torque and hence of output power.

4.12 CLOSED-LOOP OPERATION

4.12.1 Speed-Controlled Drive System

The speed-controlled dc-motor chopper drive is very similar to the phase-controlled dc-motor drive in its outer speed-control loop. The inner current loop and its control are distinctly different from those of the phase-controlled dc motor drive. This difference is due to the particular characteristics of the chopper power stage. The current loop and speed loop are examined, and their characteristics are explained, in this section. The closed-loop speed-controlled separately-excited dc motor drive is shown in Figure 4.20 for analysis, but the drive system control strategy is equally applicable to a series motor drive.

4.12.2 Current Control Loop

With inner current loop alone, the motor drive system is a torque amplifier. The commanded value of current is compared to the actual armature current, and its error is processed through a current controller. The output of the current controller, in conjunction with other constraints, determines the base drive signals of the chopper switches. The current controller can be either of the following types:

- (i) Pulse-Width-Modulation (PWM) controller
- (ii) Hysteresis controller

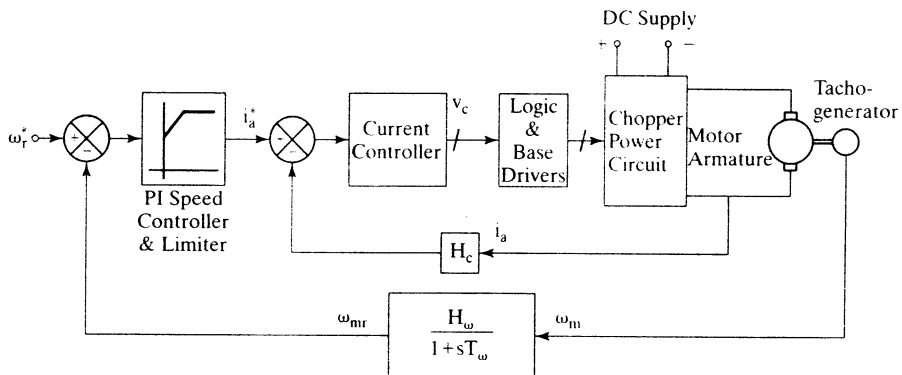


Figure 4.20 Speed-controlled dc-motor chopper drive

The selection of the current controller affects its transient response (and hence the overall speed loop bandwidth indirectly). These two controllers are described in the following sections.

4.12.3 Pulse-Width-Modulated Current Controller

The current error is fed into a controller, which could be proportional (P), proportional plus integral (PI), or proportional, integral, and differential (PID). The most commonly used controller among them is the PI controller. The current error is amplified through this controller and emerges as a control voltage, v_c . It is required to generate a proportional armature voltage from the fixed source through a chopper operation. Therefore, the control voltage is equivalent to the duty cycle of the chopper. Its realization is as follows. The control voltage is compared with a ramp signal to generate the on- and off-times, as shown in Figure 4.21. *On* signal is produced if the control voltage is greater than the ramp (carrier) signal; *off* signal is generated when the control signal is less than the ramp signal.

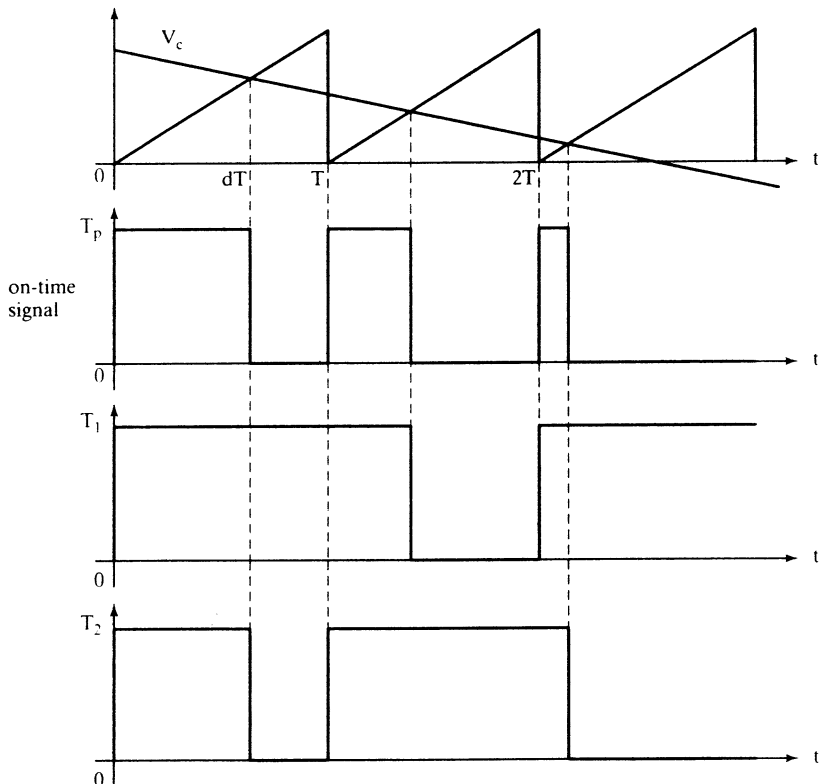


Figure 4.21 Generation of base-drive signals from current error for forward motoring when one is using the chopper shown in Figure 4.2